

# Single\_Cell\_Pipeline\_with\_ScanPy\_Cleaned

June 1, 2026

## 1 Single-Cell RNA-seq Analysis with Scanpy

This notebook walks through a **complete single-cell RNA-seq (scRNA-seq)** data analysis pipeline using the Python package **Scanpy**.

It is written for **beginners** in single-cell bioinformatics — especially those coming from a biology background.

We'll go step-by-step through:

1. Loading and preprocessing data

2. Quality control (QC)

3. Normalization and feature selection

4. Dimensionality reduction (PCA, UMAP)

5. Clustering

6. Marker gene detection and visualization

By the end of this notebook, you'll understand not only *how* to perform each step, but also *why* it's important.

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### 1.0.1 Installation

```
[ ]: !pip install scanpy
```

```
Collecting scanpy
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```
  Downloading scanpy-1.11.4-py3-none-any.whl.metadata (9.2 kB)
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Collecting anndata>=0.8 (from scanpy)
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  Downloading anndata-0.12.3-py3-none-any.whl.metadata (9.9 kB)
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Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.15.0)
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Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy) (1.5.2)
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Collecting legacy-api-wrap>=1.4.1 (from scanpy)
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  Downloading legacy_api_wrap-1.4.1-py3-none-any.whl.metadata (2.1 kB)
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Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)
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Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy) (8.4.0)
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Requirement already satisfied: networkx>=2.7.1 in
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/usr/local/lib/python3.12/dist-packages (from scanpy) (3.5)
Requirement already satisfied: numba>=0.57.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (0.60.0)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (2.0.2)
Requirement already satisfied: packaging>=21.3 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (25.0)
Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (2.2.2)
Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (1.0.1)
Requirement already satisfied: pynndescent>=0.5.13 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.13)
Requirement already satisfied: scikit-learn>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (1.6.1)
Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-
packages (from scanpy) (1.16.2)
Requirement already satisfied: seaborn>=0.13.2 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.13.2)
Collecting session-info2 (from scanpy)
  Downloading session_info2-0.2.3-py3-none-any.whl.metadata (3.4 kB)
Requirement already satisfied: statsmodels>=0.14.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from scanpy) (4.67.1)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)
Requirement already satisfied: umap-learn>=0.5.6 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)
Collecting array-api-compat>=1.7.1 (from anndata>=0.8->scanpy)
  Downloading array_api_compat-1.12.0-py3-none-any.whl.metadata (2.5 kB)
Collecting zarr!=3.0.*,>=2.18.7 (from anndata>=0.8->scanpy)
  Downloading zarr-3.1.3-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.7.5->scanpy) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy)

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(2.9.0.post0)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
/usr/local/lib/python3.12/dist-packages (from numba>=0.57.1->scanpy) (0.43.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas>=1.5.3->scanpy) (2025.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy)
(3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)
Collecting donfig>=0.8 (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading donfig-0.8.1.post1-py3-none-any.whl.metadata (5.0 kB)
Collecting numcodecs>=0.14 (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading numcodecs-0.16.3-cp312-cp312-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (3.3 kB)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (6.0.3)
Collecting crc32c>=2.7 (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy)
  Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl.metadata (7.8
kB)
Downloading scanpy-1.11.4-py3-none-any.whl (2.1 MB)
      2.1/2.1 MB
13.4 MB/s eta 0:00:00
Downloading anndata-0.12.3-py3-none-any.whl (169 kB)
      169.3/169.3 kB
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Downloading legacy_api_wrap-1.4.1-py3-none-any.whl (10.0 kB)
Downloading session_info2-0.2.3-py3-none-any.whl (16 kB)
Downloading array_api_compat-1.12.0-py3-none-any.whl (58 kB)
      58.2/58.2 kB
2.1 MB/s eta 0:00:00
Downloading zarr-3.1.3-py3-none-any.whl (276 kB)
      276.4/276.4 kB
9.8 MB/s eta 0:00:00
Downloading donfig-0.8.1.post1-py3-none-any.whl (21 kB)
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numcodecs-0.16.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (8.8
MB)
      8.8/8.8 MB
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Downloading crc32c-2.8-cp312-cp312-
manylinux1_x86_64.manylinux_2_28_x86_64.manylinux_2_5_x86_64.whl (80 kB)
      80.0/80.0 kB

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Installing collected packages: session-info2, numcodecs, legacy-api-wrap, donfig, crc32c, array-api-compat, zarr, anndata, scanpy  
Successfully installed anndata-0.12.3 array-api-compat-1.12.0 crc32c-2.8 donfig-0.8.1.post1 legacy-api-wrap-1.4.1 numcodecs-0.16.3 scanpy-1.11.4 session-info2-0.2.3 zarr-3.1.3

```
[ ]: !pip install anndata
```

Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (0.12.3)  
Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.12.0)  
Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.15.0)  
Requirement already satisfied: legacy-api-wrap in /usr/local/lib/python3.12/dist-packages (from anndata) (1.4.1)  
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata) (8.4.0)  
Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.0.2)  
Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata) (25.0)  
Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata) (2.2.2)  
Requirement already satisfied: scipy>=1.12 in /usr/local/lib/python3.12/dist-packages (from anndata) (1.16.2)  
Requirement already satisfied: zarr!=3.0.\*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata) (3.1.3)  
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata) (2025.2)  
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.\*,>=2.18.7->anndata) (0.8.1.post1)  
Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.\*,>=2.18.7->anndata) (0.16.3)  
Requirement already satisfied: typing-extensions>=4.9 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.\*,>=2.18.7->anndata) (4.15.0)  
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.\*,>=2.18.7->anndata) (6.0.3)  
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.\*,>=2.18.7->anndata) (2.8)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

```
[ ]: !pip3 install igraph
```

Collecting igraph

Downloading

igraph-0.11.9-cp39-abi3-manylinux\_2\_17\_x86\_64.manylinux2014\_x86\_64.whl.metadata (4.4 kB)

Collecting texttable>=1.6.2 (from igraph)

Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)

Downloading

igraph-0.11.9-cp39-abi3-manylinux\_2\_17\_x86\_64.manylinux2014\_x86\_64.whl (4.4 MB)  
4.4/4.4 MB

39.6 MB/s eta 0:00:00

Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)

Installing collected packages: texttable, igraph

Successfully installed igraph-0.11.9 texttable-1.7.0

```
[ ]: !pip install celltypist
```

Collecting celltypist

Downloading celltypist-1.7.1-py3-none-any.whl.metadata (45 kB)

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Requirement already satisfied: numpy>=1.19.0 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (2.0.2)

Requirement already satisfied: pandas>=1.0.5 in /usr/local/lib/python3.12/dist-packages (from celltypist) (2.2.2)

Requirement already satisfied: scikit-learn>=0.24.1 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (1.6.1)

Requirement already satisfied: openpyxl>=3.0.4 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)

Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from celltypist) (8.3.0)

Requirement already satisfied: requests>=2.23.0 in

/usr/local/lib/python3.12/dist-packages (from celltypist) (2.32.4)

Requirement already satisfied: scanpy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from celltypist) (1.11.4)

Collecting leidenalg>=0.9.0 (from celltypist)

Downloading leidenalg-0.10.2-cp38-abi3-

manylinux\_2\_17\_x86\_64.manylinux2014\_x86\_64.whl.metadata (10 kB)

Requirement already satisfied: igraph<0.12,>=0.10.0 in

/usr/local/lib/python3.12/dist-packages (from leidenalg>=0.9.0->celltypist) (0.11.9)

Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl>=3.0.4->celldtypist) (2.0.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celldtypist) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celldtypist) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.0.5->celldtypist) (2025.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celldtypist) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celldtypist) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celldtypist) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celldtypist) (2025.10.5)

Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (0.12.3)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (3.15.0)

Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (1.5.2)

Requirement already satisfied: legacy-api-wrap>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (1.4.1)

Requirement already satisfied: matplotlib>=3.7.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (3.10.0)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (3.5)

Requirement already satisfied: numba>=0.57.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (0.60.0)

Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (25.0)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (1.0.1)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (0.5.13)

Requirement already satisfied: scipy>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist) (1.16.2)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celldtypist)

(0.13.2)  
Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.2.3)  
Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.14.5)  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.67.1)  
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (4.15.0)  
Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy>=1.7.0->celltypist) (0.5.9.post2)  
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist) (3.6.0)  
Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)  
Requirement already satisfied: zarr!=3.0.\*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)  
Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)  
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)  
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)  
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)  
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)  
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)  
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)  
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)  
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)  
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-

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packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(0.8.1.post1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14-
>zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from
donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist)
(6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8-
>scanpy>=1.7.0->celltypist) (2.8)
Downloading celltypist-1.7.1-py3-none-any.whl (7.3 MB)
7.3/7.3 MB
24.2 MB/s eta 0:00:00
Downloading
leidenalg-0.10.2-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.0
MB)
2.0/2.0 MB
19.1 MB/s eta 0:00:00
Installing collected packages: leidenalg, celltypist
Successfully installed celltypist-1.7.1 leidenalg-0.10.2

```

```
[ ]: !pip install decoupler
```

```

Collecting decoupler
  Downloading decoupler-2.1.1-py3-none-any.whl.metadata (8.6 kB)
Collecting adjusttext (from decoupler)
  Downloading adjustText-1.3.0-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-
packages (from decoupler) (0.12.3)
Collecting docrep (from decoupler)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Preparing metadata (setup.py) ... done
Collecting marsilea (from decoupler)
  Downloading marsilea-0.5.6-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages
(from decoupler) (0.60.0)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from decoupler) (2.32.4)
Collecting scipy<1.16 (from decoupler)
  Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata
(61 kB)
62.0/62.0 kB
4.2 MB/s eta 0:00:00
Requirement already satisfied: session-info2 in
/usr/local/lib/python3.12/dist-packages (from decoupler) (0.2.3)

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Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from decoupler) (4.67.1)

Requirement already satisfied: numpy<2.5,>=1.23.5 in /usr/local/lib/python3.12/dist-packages (from scipy<1.16->decoupler) (2.0.2)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from adjusttext->decoupler) (3.10.0)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.12.0)

Requirement already satisfied: h5py>=3.8 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.15.0)

Requirement already satisfied: legacy-api-wrap in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (8.4.0)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (25.0)

Requirement already satisfied: pandas!=2.1.2,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (2.2.2)

Requirement already satisfied: zarr!=3.0.\*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata->decoupler) (3.1.3)

Requirement already satisfied: six in /usr/local/lib/python3.12/dist-packages (from docrep->decoupler) (1.17.0)

Collecting legendkit (from marsilea->decoupler)

  Downloading legendkit-0.3.6-py3-none-any.whl.metadata (2.6 kB)

Requirement already satisfied: platformdirs in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (4.5.0)

Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (from marsilea->decoupler) (0.13.2)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba->decoupler) (0.43.0)

Requirement already satisfied: charset\_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->decoupler) (2025.10.5)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)

```

(1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib->adjusttext->decoupler) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)
Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-
packages (from zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.8.1.post1)
Requirement already satisfied: numcodecs>=0.14 in
/usr/local/lib/python3.12/dist-packages (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (0.16.3)
Requirement already satisfied: typing-extensions>=4.9 in
/usr/local/lib/python3.12/dist-packages (from
zarr!=3.0.*,>=2.18.7->anndata->decoupler) (4.15.0)
Requirement already satisfied: pyarrow>=10.0.1 in
/usr/local/lib/python3.12/dist-packages (from
pandas[parquet]->marsilea->decoupler) (18.1.0)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
(from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)
Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-
packages (from
numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)
Downloading decoupler-2.1.1-py3-none-any.whl (117 kB)
      118.0/118.0 kB
8.6 MB/s eta 0:00:00
Downloading
scipy-1.15.3-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (37.3
MB)
      37.3/37.3 MB
33.5 MB/s eta 0:00:00
Downloading adjustText-1.3.0-py3-none-any.whl (13 kB)
Downloading marsilea-0.5.6-py3-none-any.whl (84 kB)
      85.0/85.0 kB
5.5 MB/s eta 0:00:00
Downloading legendkit-0.3.6-py3-none-any.whl (24 kB)
Building wheels for collected packages: docrep
  Building wheel for docrep (setup.py) ... done
  Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19876
sha256=2126261fe16cbe1f89506f19021566883bae6fdae763cd2f5a861013f645ad80
  Stored in directory: /root/.cache/pip/wheels/d6/19/ee/0a6a1793d91c449563b49cca
b57ce52da3e6fab7614836bd8c

```

```
Successfully built docrep
Installing collected packages: scipy, docrep, legendkit, adjusttext, marsilea,
decoupler
```

```
  Attempting uninstall: scipy
```

```
    Found existing installation: scipy 1.16.2
```

```
  Uninstalling scipy-1.16.2:
```

```
    Successfully uninstalled scipy-1.16.2
```

```
Successfully installed adjusttext-1.3.0 decoupler-2.1.1 docrep-0.3.2
legendkit-0.3.6 marsilea-0.5.6 scipy-1.15.3
```

```
[ ]: !pip install fa2-modified
```

```
Collecting fa2-modified
```

```
  Downloading fa2_modified-0.3.10.tar.gz (474 kB)
```

```
0.0/474.3
```

```
kB ? eta -:--:--
```

```
399.4/474.3 kB 11.4 MB/s eta 0:00:01
```

```
474.3/474.3 kB
```

```
6.9 MB/s eta 0:00:00
```

```
  Installing build dependencies ... done
```

```
  Getting requirements to build wheel ... done
```

```
  Preparing metadata (pyproject.toml) ... done
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (2.0.2)
```

```
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (1.15.3)
```

```
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages
(from fa2-modified) (4.67.1)
```

```
Building wheels for collected packages: fa2-modified
```

```
  Building wheel for fa2-modified (pyproject.toml) ... done
```

```
  Created wheel for fa2-modified:
```

```
filename=fa2_modified-0.3.10-cp312-cp312-linux_x86_64.whl size=493566
```

```
sha256=a69ec6fb4fb2f5bb2781f699e765b0b0c65d9095abb4de6ef732f7f7d91a014c
```

```
  Stored in directory: /root/.cache/pip/wheels/e7/b5/c2/e87afecc23dc78723045cde9
a733677ee2b3c1374a7ed3a5e8
```

```
Successfully built fa2-modified
```

```
Installing collected packages: fa2-modified
```

```
Successfully installed fa2-modified-0.3.10
```

## 1.0.2 Loading Data

In this step, we load the single-cell expression matrix into an **AnnData** object — the core data structure in Scanpy. It contains: - **adata.X**: the expression matrix (cells × genes) - **adata.obs**: metadata for each cell - **adata.var**: metadata for each gene

```
[ ]: #Import core single cell tools
```

```
import scanpy as sc
import anndata as ad
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/_utils/__init__.py:33:
FutureWarning: `__version__` is deprecated, use
`importlib.metadata.version('anndata')` instead.
    from anndata import __version__ as anndata_version
/usr/local/lib/python3.12/dist-packages/scanpy/__init__.py:24: FutureWarning:
`__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
    if Version(anndata.__version__) >= Version("0.11.0rc2"):
/usr/local/lib/python3.12/dist-packages/scanpy/readwrite.py:16: FutureWarning:
`__version__` is deprecated, use `importlib.metadata.version('anndata')`
instead.
    if Version(anndata.__version__) >= Version("0.11.0rc2"):
```

```
[ ]: !wget https://cf.10xgenomics.com/samples/cell-vdj/8.0.1/
      ↪10k_5p_Human_diseased_PBMC_ALL_Fresh/
      ↪10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5
```

```
--2025-10-20 13:25:00-- https://cf.10xgenomics.com/samples/cell-vdj/8.0.1/10k_5
p_Human_diseased_PBMC_ALL_Fresh/10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filde
red_feature_bc_matrix.h5
Resolving cf.10xgenomics.com (cf.10xgenomics.com)... 104.18.0.173, 104.18.1.173,
2606:4700::6812:1ad, ...
Connecting to cf.10xgenomics.com (cf.10xgenomics.com)|104.18.0.173|:443...
connected.
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 65402953 (62M) [application/x-hdf5]
```

```
Saving to:
```

```
'10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5'
```

```
10k_5p_Human_diseas 100%[=====>] 62.37M 37.2MB/s in 1.7s
```

```
2025-10-20 13:25:02 (37.2 MB/s) -
```

```
'10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.h5' saved
[65402953/65402953]
```

```
[ ]: fresh_blood_adata = sc.
```

```
    ↪read_10x_h5('10k_5p_Human_diseased_PBMC_ALL_Fresh_count_filtered_feature_bc_matrix.
    ↪h5')
```

```
/usr/local/lib/python3.12/dist-packages/anndata/_core/anndata.py:1794:
```

```
UserWarning: Variable names are not unique. To make them unique, call
`.var_names_make_unique`.
```

```

utils.warn_names_duplicates("var")
/usr/local/lib/python3.12/dist-packages/anndata/_core/anndata.py:1794:
UserWarning: Variable names are not unique. To make them unique, call
`.var_names_make_unique`.
utils.warn_names_duplicates("var")

```

```
[ ]: print(fresh_blood_adata)
```

```

AnnData object with n_obs × n_vars = 13853 × 38606
    var: 'gene_ids', 'feature_types', 'genome'

```

```
[ ]: # the dimensions of our dataset
fresh_blood_adata.shape
```

```
[ ]: (13439, 9650)
```

```
[ ]: #13853 cells
#38606 genes
```

```
[ ]: # let's look at the first 5 rows describing the genes in our dataset
fresh_blood_adata.var.head()
```

```
[ ]:
           gene_ids  feature_types  genome
DDX11L2      ENSG00000290825  Gene Expression  GRCh38
MIR1302-2HG   ENSG00000243485  Gene Expression  GRCh38
FAM138A       ENSG00000237613  Gene Expression  GRCh38
ENSG00000290826 ENSG00000290826  Gene Expression  GRCh38
OR4F5         ENSG00000186092  Gene Expression  GRCh38
```

```
[ ]: # let's look at the first 5 rows describing the cells (ID) in our dataset
fresh_blood_adata.obs.head()
```

```
[ ]: Empty DataFrame
Columns: []
Index: [AAACCAAAGAACAGAC-1, AAACCAAAGCAAGATA-1, AAACCAAAGCTGGTTA-1,
AAACCAGCACCTAACG-1, AAACCAGCACGCGTTA-1]
```

```
[ ]: # How about both. in a proper dataframe format

fresh_blood_adata.to_df()
```

```
[ ]:
           DDX11L2  MIR1302-2HG  FAM138A  ENSG00000290826  OR4F5  \
AAACCAAAGAACAGAC-1    0.0         0.0         0.0         0.0    0.0
AAACCAAAGCAAGATA-1    0.0         0.0         0.0         0.0    0.0
AAACCAAAGCTGGTTA-1    0.0         0.0         0.0         0.0    0.0
AAACCAGCACCTAACG-1    0.0         0.0         0.0         0.0    0.0
AAACCAGCACGCGTTA-1    0.0         0.0         0.0         0.0    0.0
...               ...         ...         ...         ...    ...
```

|                    |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|
| GTTGTCTTCGTCCATT-2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| GTTGTGCAGAAGCCGA-2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| GTTGTGCAGATTACAC-2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| GTTGTGGGTAATGGGA-2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| GTTGTGGGTCGTCAGA-2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

|                    |                 |                 |                 |   |
|--------------------|-----------------|-----------------|-----------------|---|
|                    | ENSG00000238009 | ENSG00000239945 | ENSG00000239906 | \ |
| AAACCAAAGAACAGAC-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAAAGCAAGATA-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAAAGCTGGTTA-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAGCACCTAACG-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAGCACGCGTTA-1 | 0.0             | 0.0             | 0.0             |   |
| ...                | ...             | ...             | ...             |   |
| GTTGTCTTCGTCCATT-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGCAGAAGCCGA-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGCAGATTACAC-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGGGTAATGGGA-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGGGTCGTCAGA-2 | 0.0             | 0.0             | 0.0             |   |

|                    |                 |                 |     |                 |   |
|--------------------|-----------------|-----------------|-----|-----------------|---|
|                    | ENSG00000241860 | ENSG00000241599 | ... | ENSG00000275249 | \ |
| AAACCAAAGAACAGAC-1 | 0.0             | 0.0             | ... | 0.0             |   |
| AAACCAAAGCAAGATA-1 | 0.0             | 0.0             | ... | 0.0             |   |
| AAACCAAAGCTGGTTA-1 | 0.0             | 0.0             | ... | 0.0             |   |
| AAACCAGCACCTAACG-1 | 0.0             | 0.0             | ... | 0.0             |   |
| AAACCAGCACGCGTTA-1 | 0.0             | 0.0             | ... | 0.0             |   |
| ...                | ...             | ...             | ... | ...             |   |
| GTTGTCTTCGTCCATT-2 | 0.0             | 0.0             | ... | 0.0             |   |
| GTTGTGCAGAAGCCGA-2 | 0.0             | 0.0             | ... | 0.0             |   |
| GTTGTGCAGATTACAC-2 | 0.0             | 0.0             | ... | 0.0             |   |
| GTTGTGGGTAATGGGA-2 | 0.0             | 0.0             | ... | 0.0             |   |
| GTTGTGGGTCGTCAGA-2 | 0.0             | 0.0             | ... | 0.0             |   |

|                    |                 |                 |                 |   |
|--------------------|-----------------|-----------------|-----------------|---|
|                    | ENSG00000274792 | ENSG00000274175 | ENSG00000275869 | \ |
| AAACCAAAGAACAGAC-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAAAGCAAGATA-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAAAGCTGGTTA-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAGCACCTAACG-1 | 0.0             | 0.0             | 0.0             |   |
| AAACCAGCACGCGTTA-1 | 0.0             | 0.0             | 0.0             |   |
| ...                | ...             | ...             | ...             |   |
| GTTGTCTTCGTCCATT-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGCAGAAGCCGA-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGCAGATTACAC-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGGGTAATGGGA-2 | 0.0             | 0.0             | 0.0             |   |
| GTTGTGGGTCGTCAGA-2 | 0.0             | 0.0             | 0.0             |   |

|                    |                 |                 |                 |   |
|--------------------|-----------------|-----------------|-----------------|---|
|                    | ENSG00000273554 | ENSG00000277836 | ENSG00000278633 | \ |
| AAACCAAAGAACAGAC-1 | 0.0             | 0.0             | 0.0             |   |

|                    |     |     |     |
|--------------------|-----|-----|-----|
| AAACCAAAGCAAGATA-1 | 0.0 | 0.0 | 0.0 |
| AAACCAAAGCTGGTTA-1 | 0.0 | 0.0 | 0.0 |
| AAACCAGCACCTAACG-1 | 0.0 | 0.0 | 0.0 |
| AAACCAGCACGCGTTA-1 | 0.0 | 0.0 | 0.0 |
| ...                | ... | ... | ... |
| GTTGTCTTCGTCCATT-2 | 0.0 | 0.0 | 0.0 |
| GTTGTGCAGAAGCCGA-2 | 0.0 | 0.0 | 0.0 |
| GTTGTGCAGATTACAC-2 | 0.0 | 0.0 | 0.0 |
| GTTGTGGGTAATGGGA-2 | 0.0 | 0.0 | 0.0 |
| GTTGTGGGTCGTCAGA-2 | 0.0 | 0.0 | 0.0 |

|                    | ENSG00000276017 | ENSG00000278817 | ENSG00000277196 |
|--------------------|-----------------|-----------------|-----------------|
| AAACCAAAGAACAGAC-1 | 0.0             | 0.0             | 0.0             |
| AAACCAAAGCAAGATA-1 | 0.0             | 0.0             | 0.0             |
| AAACCAAAGCTGGTTA-1 | 0.0             | 0.0             | 0.0             |
| AAACCAGCACCTAACG-1 | 0.0             | 0.0             | 0.0             |
| AAACCAGCACGCGTTA-1 | 0.0             | 0.0             | 0.0             |
| ...                | ...             | ...             | ...             |
| GTTGTCTTCGTCCATT-2 | 0.0             | 0.0             | 0.0             |
| GTTGTGCAGAAGCCGA-2 | 0.0             | 0.0             | 0.0             |
| GTTGTGCAGATTACAC-2 | 0.0             | 0.0             | 0.0             |
| GTTGTGGGTAATGGGA-2 | 0.0             | 0.0             | 0.0             |
| GTTGTGGGTCGTCAGA-2 | 0.0             | 0.0             | 0.0             |

[13853 rows x 38606 columns]

### 1.0.3 Quality Control (QC)

QC ensures we only keep high-quality cells and informative genes. Typical filters remove: - Harmonize unique gene names (avoid gene duplications from old pipelines) - Cells with too few genes (likely dead) - Cells with too many genes (possible doublets) - Genes expressed in very few cells (uninformative)

```
[ ]: # A useful step for older datasets
fresh_blood_adata.var_names_make_unique()
fresh_blood_adata.obs_names_make_unique()
```

```
[ ]: #Let's search for possible contamination from dying cells, ribosomal
      ↳transcripts or hemoglobin

#Cells with a high proportion of mitochondrial reads (say >10-20%) are likely
      ↳stressed, apoptotic, or poorly captured
#Ribosomal transcripts are removed because they represent global
      ↳transcriptional activity, not cell-type-specific biology
#Instead of true cell populations, high HB signal often represents ambient RNA
      ↳contamination from lysed red blood cells
```

```

fresh_blood_adata.var['MT'] = fresh_blood_adata.var_names.str.startswith("MT-")
fresh_blood_adata.var['RIBO'] = fresh_blood_adata.var_names.str.
    ↳startswith("RPS", "RPL")
fresh_blood_adata.var['HB'] = fresh_blood_adata.var_names.str.
    ↳startswith("^HB[^(P)]")

```

```
[ ]: #let's just take a quick look at one of them.
```

```

mt_genes = fresh_blood_adata.var[fresh_blood_adata.var['MT']]
mt_genes

```

```
[ ]:
```

|         | gene_ids        | feature_types   | genome | MT   | RIBO  | HB    |
|---------|-----------------|-----------------|--------|------|-------|-------|
| MT-ND1  | ENSG00000198888 | Gene Expression | GRCh38 | True | False | False |
| MT-ND2  | ENSG00000198763 | Gene Expression | GRCh38 | True | False | False |
| MT-CO1  | ENSG00000198804 | Gene Expression | GRCh38 | True | False | False |
| MT-CO2  | ENSG00000198712 | Gene Expression | GRCh38 | True | False | False |
| MT-ATP8 | ENSG00000228253 | Gene Expression | GRCh38 | True | False | False |
| MT-ATP6 | ENSG00000198899 | Gene Expression | GRCh38 | True | False | False |
| MT-CO3  | ENSG00000198938 | Gene Expression | GRCh38 | True | False | False |
| MT-ND3  | ENSG00000198840 | Gene Expression | GRCh38 | True | False | False |
| MT-ND4L | ENSG00000212907 | Gene Expression | GRCh38 | True | False | False |
| MT-ND4  | ENSG00000198886 | Gene Expression | GRCh38 | True | False | False |
| MT-ND5  | ENSG00000198786 | Gene Expression | GRCh38 | True | False | False |
| MT-ND6  | ENSG00000198695 | Gene Expression | GRCh38 | True | False | False |
| MT-CYB  | ENSG00000198727 | Gene Expression | GRCh38 | True | False | False |

```
[ ]: #calculate the qc metrics
```

```

sc.pp.calculate_qc_metrics(
    fresh_blood_adata, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)

```

```
[ ]: #note that it is also included in the headers of obs
```

```
fresh_blood_adata.obs.head()
```

```
[ ]:
```

|                    | n_genes_by_counts | log1p_n_genes_by_counts | total_counts | \ |
|--------------------|-------------------|-------------------------|--------------|---|
| AAACCAAAGAACAGAC-1 | 3351              | 8.117312                | 9437.0       |   |
| AAACCAAAGCAAGATA-1 | 2718              | 7.908019                | 9637.0       |   |
| AAACCAAAGCTGGTTA-1 | 3262              | 8.090402                | 10236.0      |   |
| AAACCAGCACCTAACG-1 | 3184              | 8.066208                | 9737.0       |   |
| AAACCAGCACGCGTTA-1 | 3342              | 8.114624                | 11524.0      |   |

|                    | log1p_total_counts | pct_counts_in_top_50_genes | \ |
|--------------------|--------------------|----------------------------|---|
| AAACCAAAGAACAGAC-1 | 9.152499           | 22.719084                  |   |
| AAACCAAAGCAAGATA-1 | 9.173469           | 35.872159                  |   |



|                    |          |           |
|--------------------|----------|-----------|
| AAACCAAAGCTGGTTA-1 | 9.233764 | 29.054318 |
| AAACCAGCACCTAACG-1 | 9.183791 | 26.887132 |
| AAACCAGCACGCGTTA-1 | 9.352274 | 28.089205 |

|                    | pct_counts_in_top_100_genes | pct_counts_in_top_200_genes \ |
|--------------------|-----------------------------|-------------------------------|
| AAACCAAAGAACAGAC-1 | 30.772491                   | 41.178341                     |
| AAACCAAAGCAAGATA-1 | 49.175054                   | 58.285774                     |
| AAACCAAAGCTGGTTA-1 | 39.576006                   | 49.267292                     |
| AAACCAGCACCTAACG-1 | 36.664270                   | 47.047345                     |
| AAACCAGCACGCGTTA-1 | 40.758417                   | 51.292954                     |

|                    | pct_counts_in_top_500_genes | total_counts_MT \ |
|--------------------|-----------------------------|-------------------|
| AAACCAAAGAACAGAC-1 | 56.575183                   | 420.0             |
| AAACCAAAGCAAGATA-1 | 69.959531                   | 351.0             |
| AAACCAAAGCTGGTTA-1 | 62.289957                   | 472.0             |
| AAACCAGCACCTAACG-1 | 60.778474                   | 349.0             |
| AAACCAGCACGCGTTA-1 | 64.213815                   | 501.0             |

|                    | loglp_total_counts_MT | pct_counts_MT | total_counts_RIBO \ |
|--------------------|-----------------------|---------------|---------------------|
| AAACCAAAGAACAGAC-1 | 6.042633              | 4.450567      | 479.0               |
| AAACCAAAGCAAGATA-1 | 5.863631              | 3.642212      | 1455.0              |
| AAACCAAAGCTGGTTA-1 | 6.159095              | 4.611176      | 1048.0              |
| AAACCAGCACCTAACG-1 | 5.857933              | 3.584266      | 696.0               |
| AAACCAGCACGCGTTA-1 | 6.218600              | 4.347449      | 1302.0              |

|                    | loglp_total_counts_RIBO | pct_counts_RIBO | total_counts_HB \ |
|--------------------|-------------------------|-----------------|-------------------|
| AAACCAAAGAACAGAC-1 | 6.173786                | 5.075766        | 0.0               |
| AAACCAAAGCAAGATA-1 | 7.283448                | 15.098059       | 0.0               |
| AAACCAAAGCTGGTTA-1 | 6.955593                | 10.238374       | 0.0               |
| AAACCAGCACCTAACG-1 | 6.546785                | 7.147993        | 0.0               |
| AAACCAGCACGCGTTA-1 | 7.172425                | 11.298161       | 0.0               |

|                    | loglp_total_counts_HB | pct_counts_HB |
|--------------------|-----------------------|---------------|
| AAACCAAAGAACAGAC-1 | 0.0                   | 0.0           |
| AAACCAAAGCAAGATA-1 | 0.0                   | 0.0           |
| AAACCAAAGCTGGTTA-1 | 0.0                   | 0.0           |
| AAACCAGCACCTAACG-1 | 0.0                   | 0.0           |
| AAACCAGCACGCGTTA-1 | 0.0                   | 0.0           |

```
[ ]: #and your gene list
fresh_blood_adata.var.head()
```

```
[ ]:
      gene_ids  feature_types  genome  MT  RIBO \
DDX11L2  ENSG00000290825  Gene Expression  GRCh38  False  False
MIR1302-2HG  ENSG00000243485  Gene Expression  GRCh38  False  False
FAM138A  ENSG00000237613  Gene Expression  GRCh38  False  False
ENSG00000290826  ENSG00000290826  Gene Expression  GRCh38  False  False
```

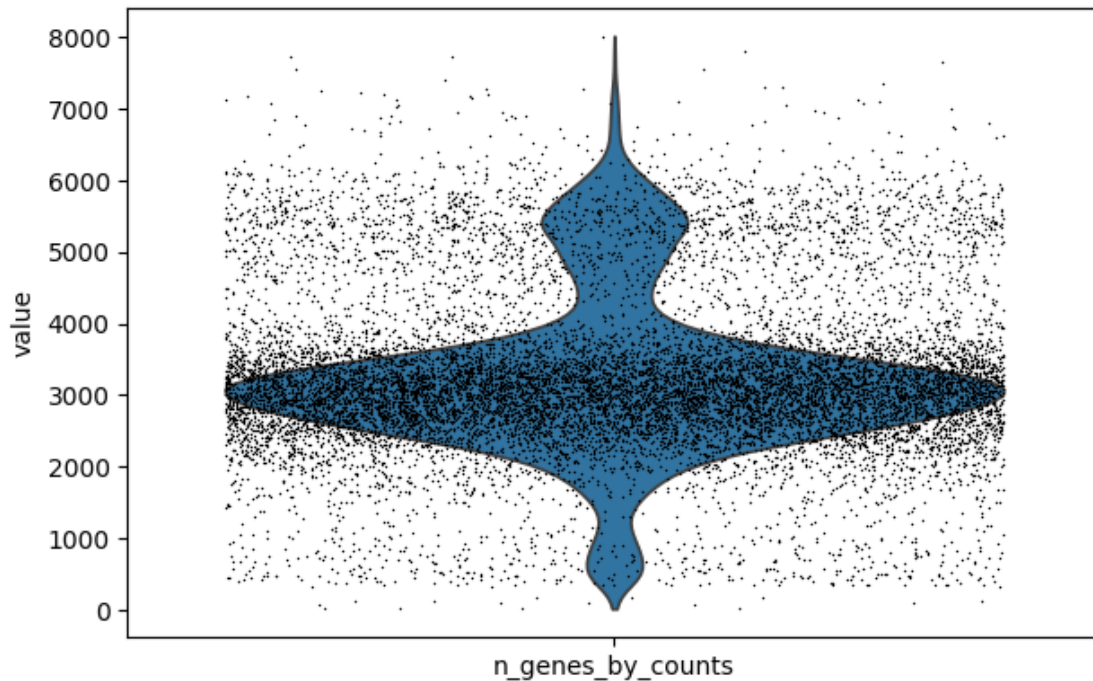
OR4F5                      ENSG00000186092   Gene Expression   GRCh38   False   False

|                 | HB    | n_cells_by_counts | mean_counts | log1p_mean_counts | \ |
|-----------------|-------|-------------------|-------------|-------------------|---|
| DDX11L2         | False | 7                 | 0.000505    | 0.000505          |   |
| MIR1302-2HG     | False | 1                 | 0.000072    | 0.000072          |   |
| FAM138A         | False | 0                 | 0.000000    | 0.000000          |   |
| ENSG00000290826 | False | 0                 | 0.000000    | 0.000000          |   |
| OR4F5           | False | 0                 | 0.000000    | 0.000000          |   |

|                 | pct_dropout_by_counts | total_counts | log1p_total_counts |
|-----------------|-----------------------|--------------|--------------------|
| DDX11L2         | 99.949469             | 7.0          | 2.079442           |
| MIR1302-2HG     | 99.992781             | 1.0          | 0.693147           |
| FAM138A         | 100.000000            | 0.0          | 0.000000           |
| ENSG00000290826 | 100.000000            | 0.0          | 0.000000           |
| OR4F5           | 100.000000            | 0.0          | 0.000000           |

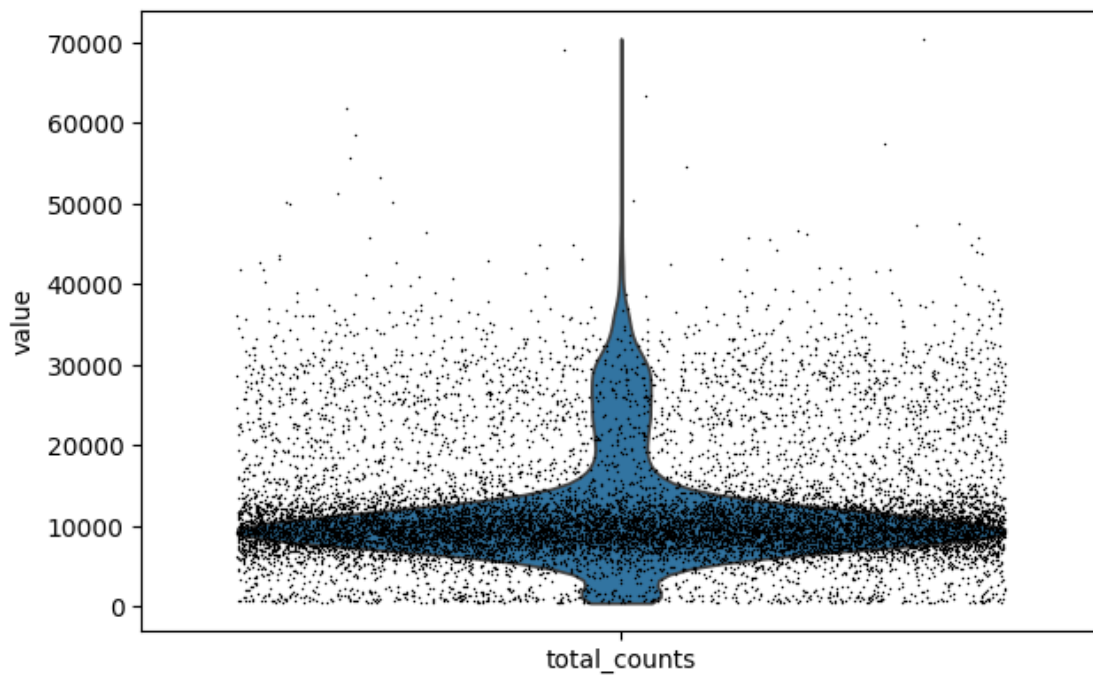
```
[ ]: #what is the average number of genes that have at least one detected identifier
      ↪ in each cell.
      #in other words, the number of genes expressed in each cell

sc.pl.violin(
    fresh_blood_adata,
    ["n_genes_by_counts"],
    jitter=0.4,
    multi_panel=False,
)
```

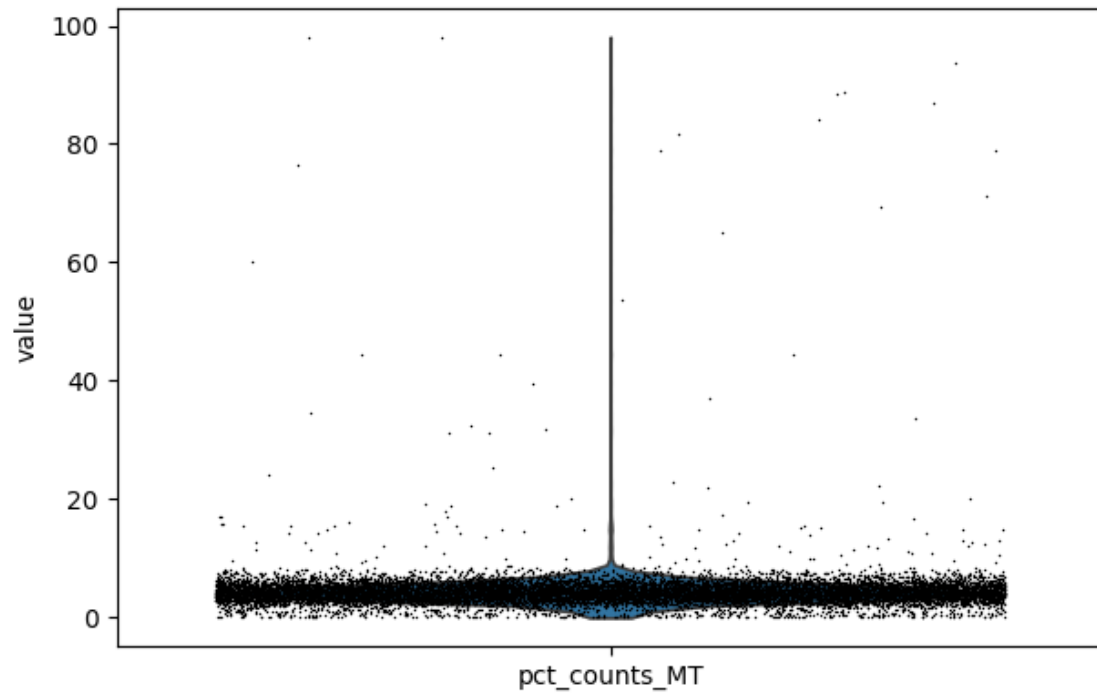


```
[ ]: #What is the total number of molecules (UMI) detected in a cell.  
#basically you can have 10 UMI molecules but they are all pointing to the same  
↳ gene.
```

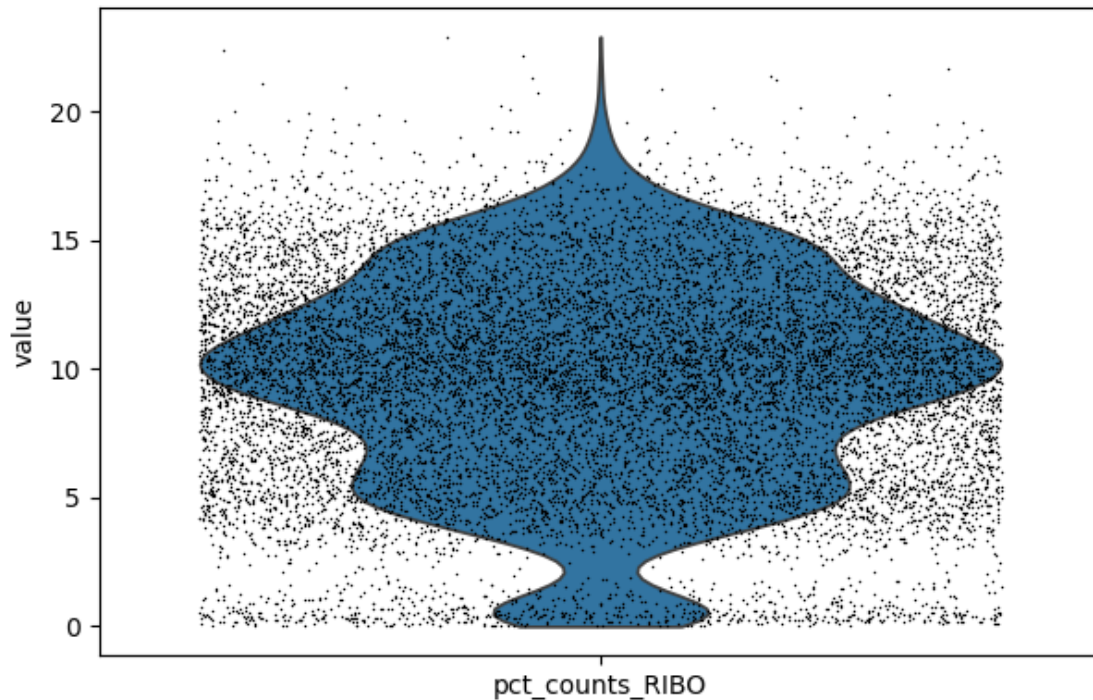
```
sc.pl.violin(  
    fresh_blood_adata,  
    ["total_counts"],  
    jitter=0.4,  
    multi_panel=False,  
)
```



```
[ ]: #what about those mitochondrial genes?  
sc.pl.violin(  
    fresh_blood_adata,  
    ["pct_counts_MT"],  
    jitter=0.4,  
    multi_panel=False,  
)
```

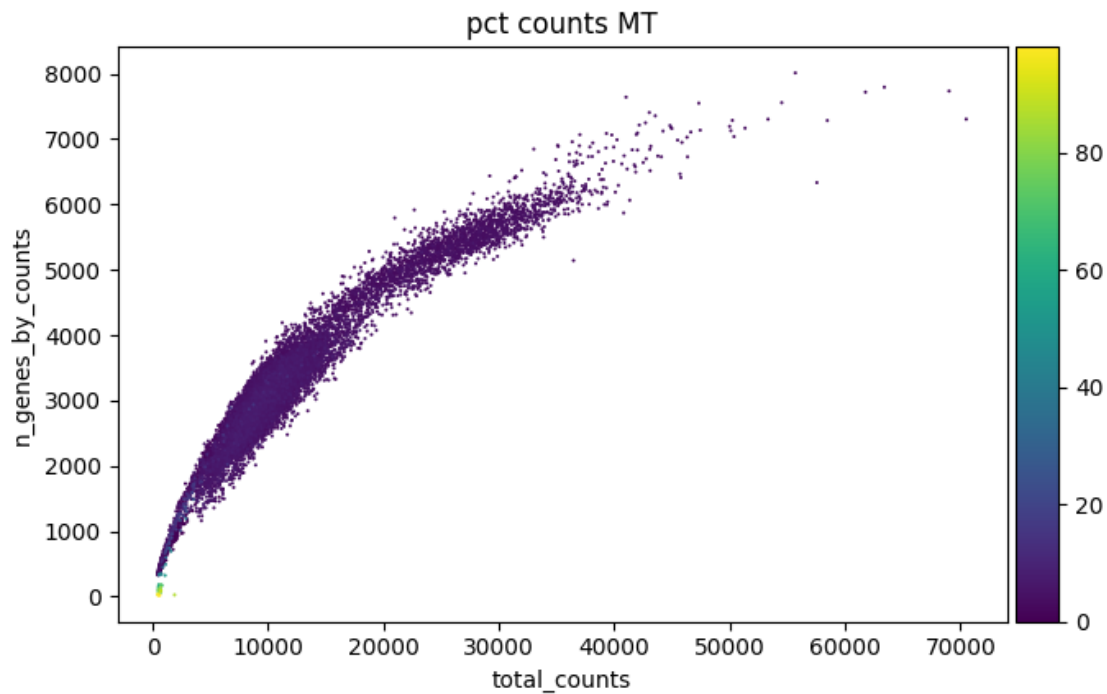


```
[ ]: #and the ribosomal genes
sc.pl.violin(
    fresh_blood_adata,
    ["pct_counts_RIBO"],
    jitter=0.4,
    multi_panel=False,
)
```



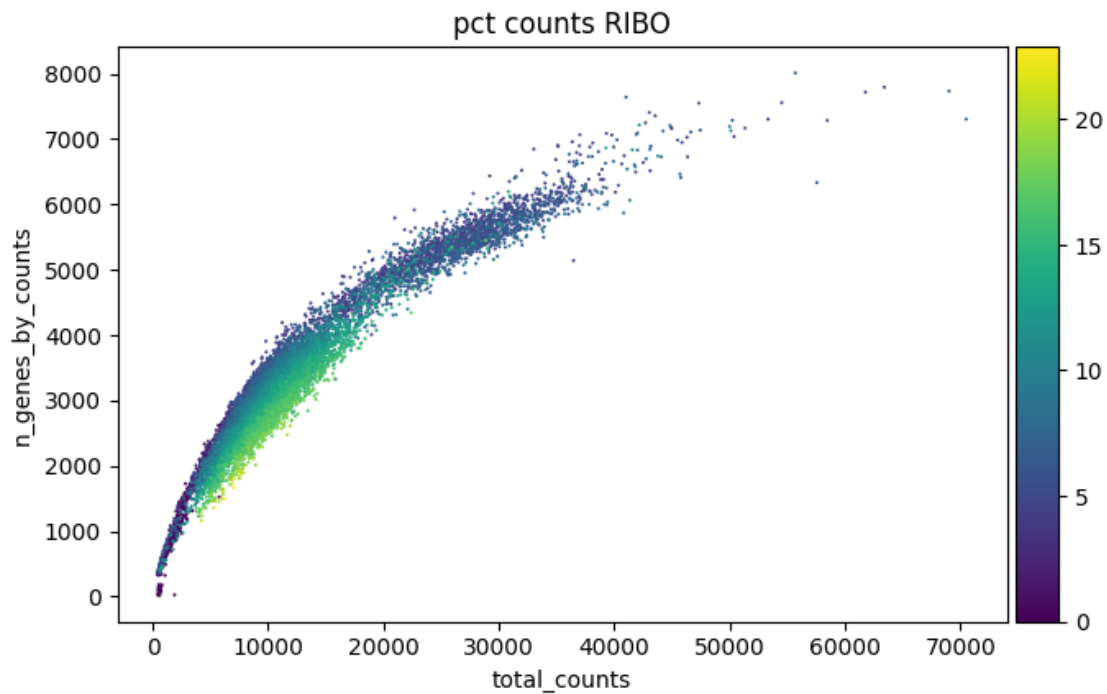
```
[ ]: #let's visualize the three of them. And let's see where the mitochondrial genes
      ↪ are
      sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
      ↪ color="pct_counts_MT")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



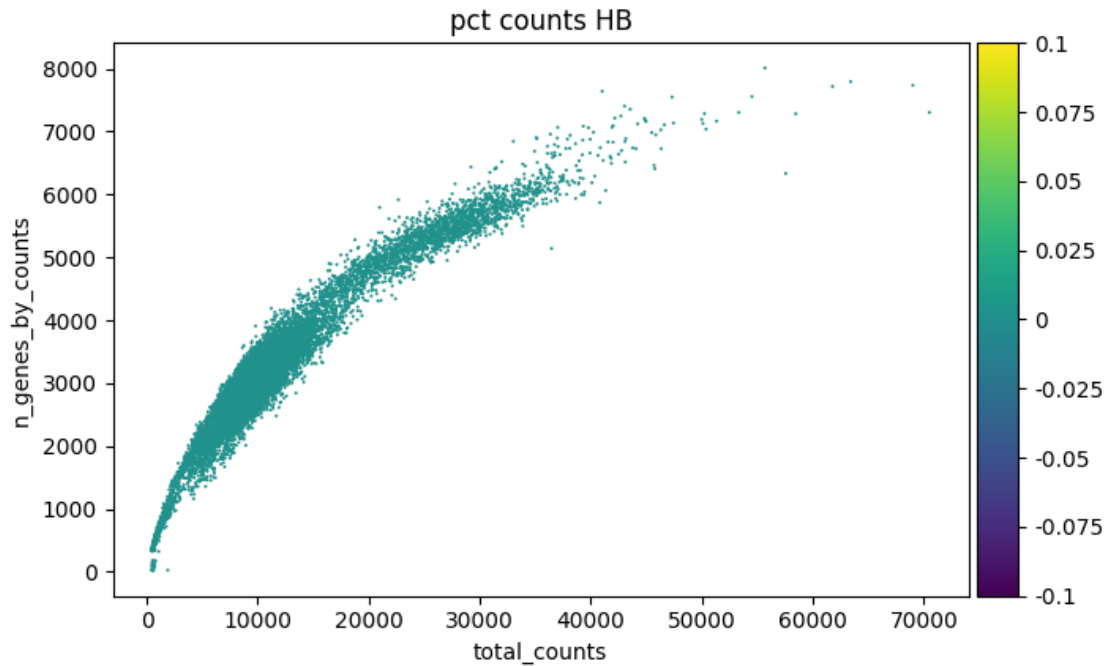
```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_RIB0")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
                  color="pct_counts_HB")
```

```
/usr/local/lib/python3.12/dist-packages/scanpy/plotting/_anndata.py:397:
FutureWarning: Use obs (e.g. `k in adata.obs` or
`str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys, AnnData.obs_keys
is deprecated and will be removed in the future.
    if key in adata.obs_keys():
```



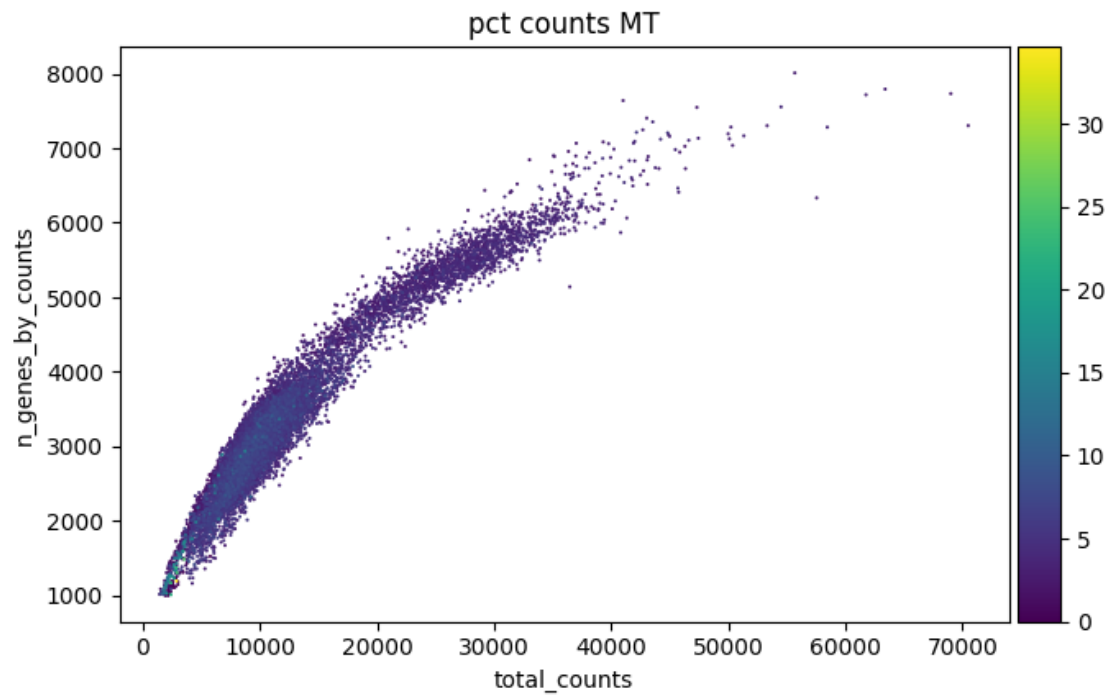
```
[ ]: """
    Additionally, it is important to note that for datasets with multiple batches,
    quality control should be performed for each sample individually as quality
    control thresholds can vary substantially between batches.
    """
```

```
[ ]: '\nAdditionally, it is important to note that for datasets with multiple
batches,\nquality control should be performed for each sample individually as
quality\ncontrol thresholds can vary substantially between batches.\n'
```

```
[ ]: #Let's use the data MT plot to select things to remove
sc.pp.filter_cells(fresh_blood_adata, min_genes=1000)
sc.pp.filter_genes(fresh_blood_adata, min_cells=1000)
```

```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",
    ↪color="pct_counts_MT")
```

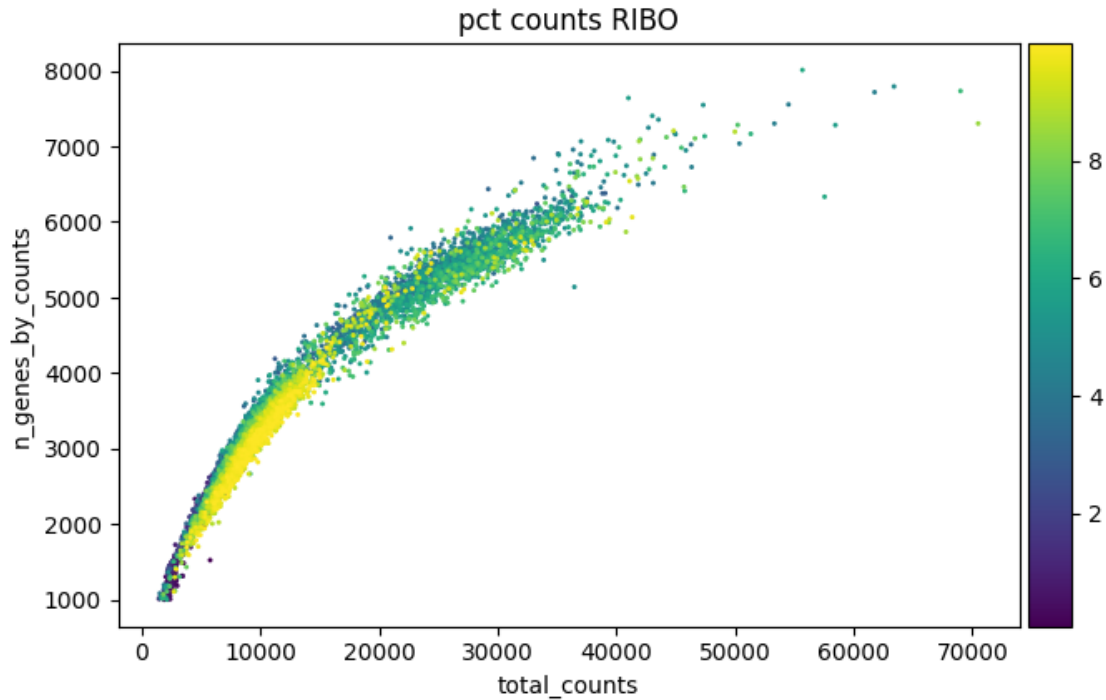




```
[ ]: # we can also further filter for ribosomal contaminations using
```

```
fresh_blood_adata = fresh_blood_adata[  
    fresh_blood_adata.obs['pct_counts_RIBO'] < 10,  
    :  
]
```

```
[ ]: sc.pl.scatter(fresh_blood_adata, "total_counts", "n_genes_by_counts",  
    ↪color="pct_counts_RIBO")
```



```
[ ]: #doublet detection
    ##Identifying doublets is crucial as they can lead to misclassifications or
    ##distortions in downstream analysis steps
```

```
[ ]: sc.pp.scrublet(fresh_blood_adata) #if you have multiple batch samples, you can
    ↪ specify it with batch_key="sample"
```

WARNING: adata.X seems to be already log-transformed.

#### 1.0.4 Normalization

Normalization adjusts for sequencing depth differences between cells. Here, we scale counts so each cell has the same total expression level.

```
[ ]: #Normalization
```

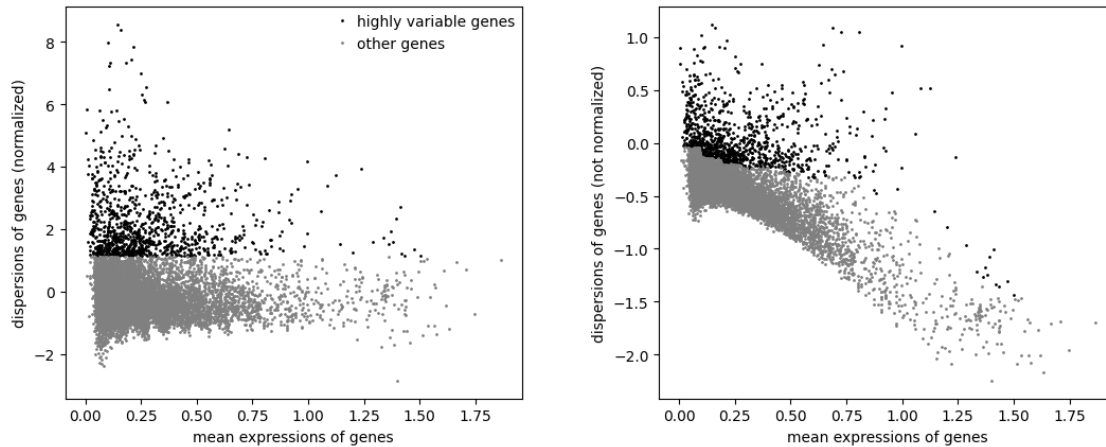
```
[ ]: # Save a copy of the data
fresh_blood_adata.layers["counts"] = fresh_blood_adata.X.copy()
```

```
[ ]: # Normalizing to median total counts
sc.pp.normalize_total(fresh_blood_adata)
# Logarithmize the data
sc.pp.log1p(fresh_blood_adata)
```

WARNING: adata.X seems to be already log-transformed.

```
[ ]: #Feature selection  
#selecting the top 1000 most variable genes  
sc.pp.highly_variable_genes(fresh_blood_adata, n_top_genes=1000)
```

```
[ ]: sc.pl.highly_variable_genes(fresh_blood_adata )  
#left is normalized  
#right is not
```



### 1.0.5 Dimensionality Reduction (PCA)

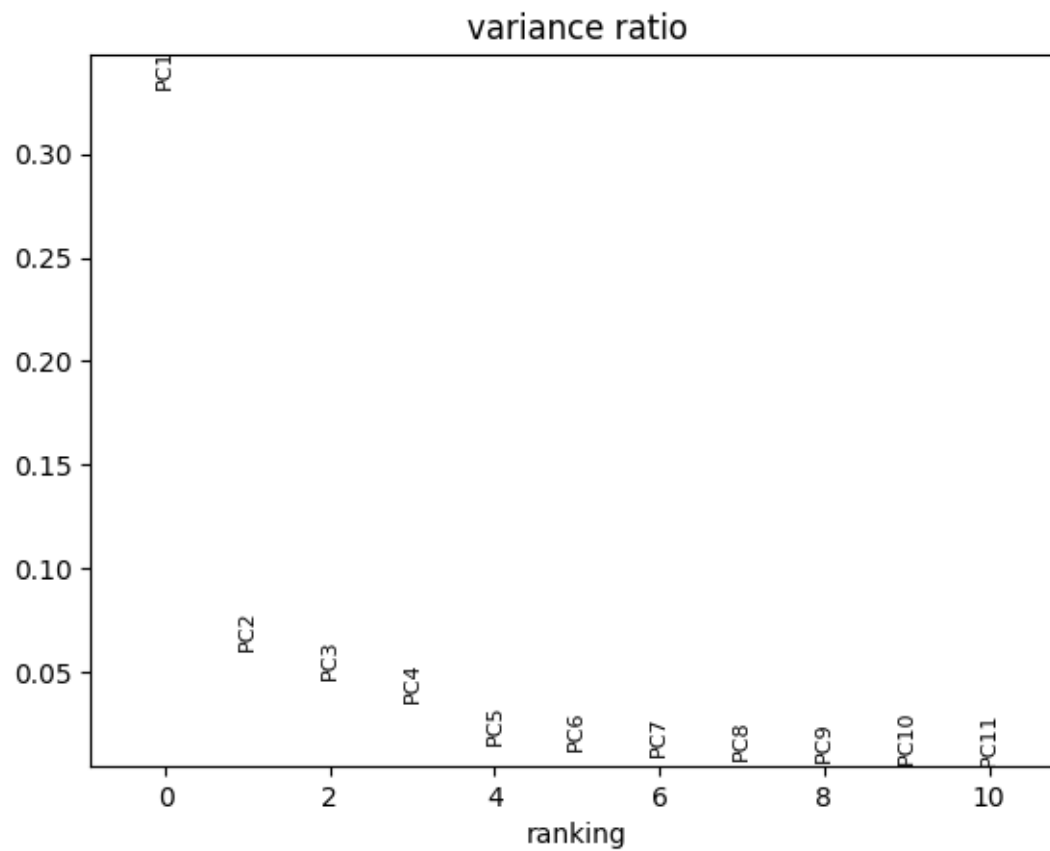
We use **Principal Component Analysis (PCA)** to reduce data complexity and highlight key variation patterns. This makes later steps like clustering and visualization faster and more robust.

In single-cell RNA-seq, each cell has expression values for thousands of genes, creating a huge, noisy matrix. PCA compresses this high-dimensional data into a smaller set of features (typically 30–50 components) that summarize the key biological and technical variation across cells.

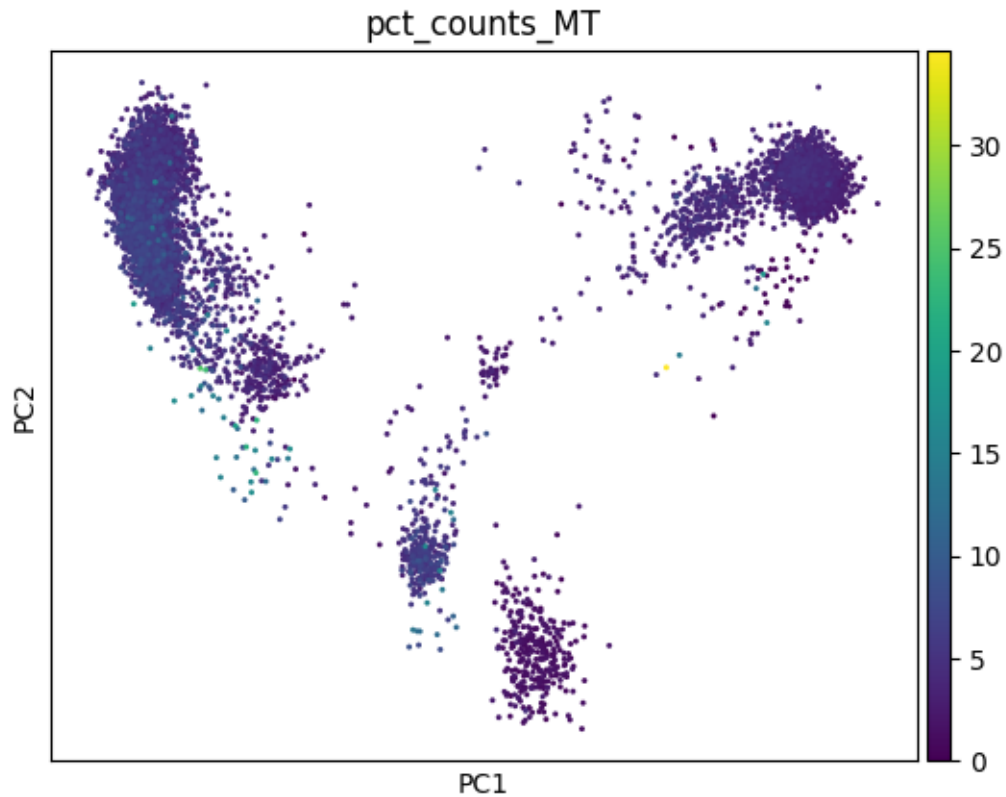
- Noise reduction: scRNA-seq data are sparse and noisy. PCA focuses on the strongest correlated gene expression patterns, discarding random noise.
- Computational efficiency: Downstream analyses like clustering, UMAP, or t-SNE run much faster and more robustly on 30 PCs than on 20,000 genes.
- Signal extraction: The top PCs often correspond to meaningful biological structure—cell type, cell cycle state, or activation level—while later PCs capture less relevant variation.

```
[ ]: sc.tl.pca(fresh_blood_adata)
```

```
[ ]: sc.pl.pca_variance_ratio(fresh_blood_adata, n_pcs=10, log=False)
```



```
[ ]: sc.pl.pca(  
    fresh_blood_adata,  
    color=["pct_counts_MT"]  
)
```

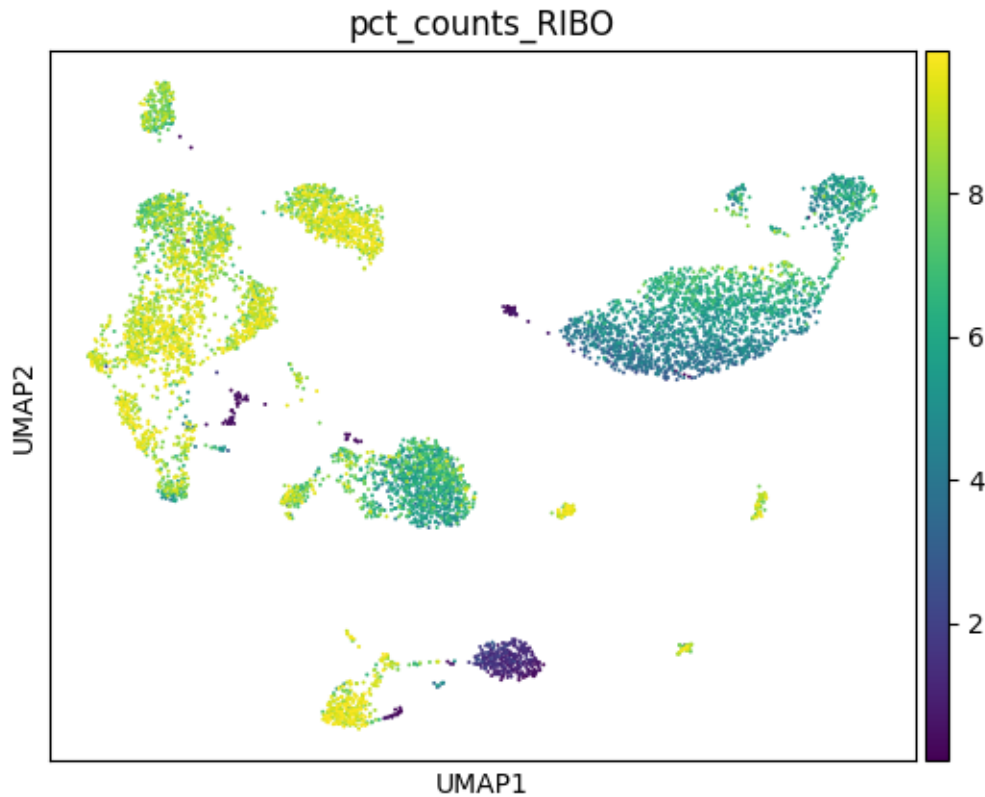


```
[ ]: ## Nearest Neighbour
      # Let us compute the neighborhood graph of cells using the PCA representation
      ↪ of the data matrix.
      # basically we want to cluster the PCA components
```

```
[ ]: sc.pp.neighbors(fresh_blood_adata)
```

```
[ ]: sc.tl.umap(fresh_blood_adata)
```

```
[ ]: sc.pl.umap(
      fresh_blood_adata,
      color=["pct_counts_RIBO"],
      size=8,
    )
```



```
[ ]: ## Clustering by communities.

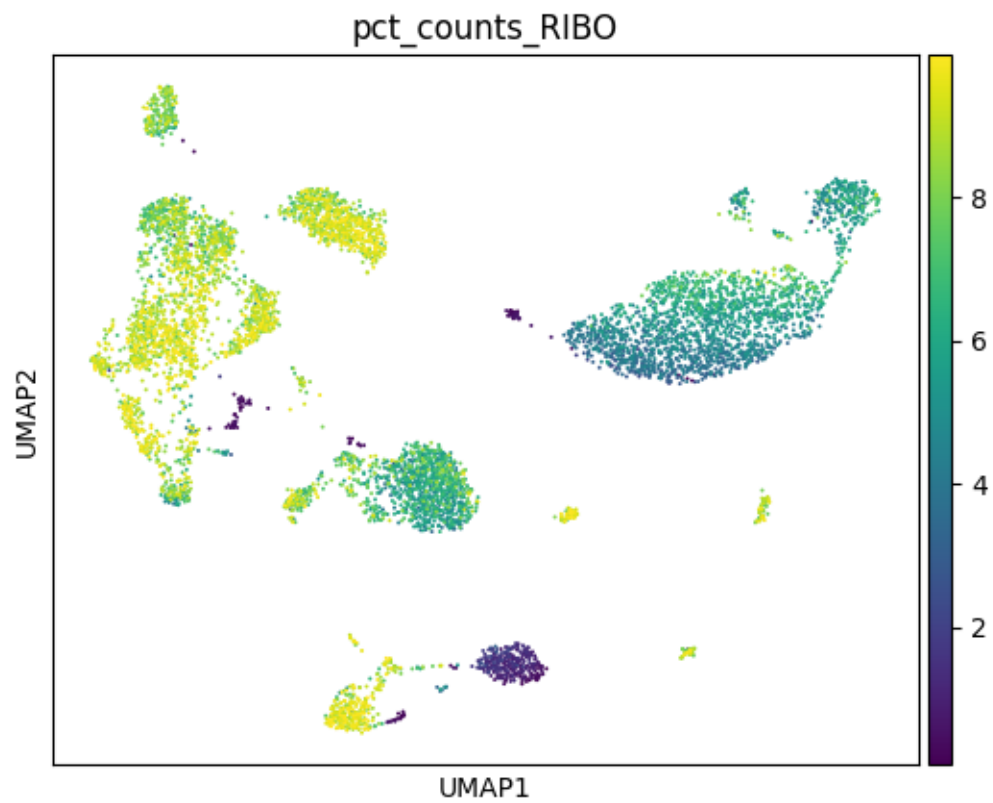
##Clustering by communities in single-cell RNA-seq is the process of grouping
    ↳ cells that show similar expression profiles - essentially, discovering
    ↳ putative cell types or states.

## Once PCA compresses your data into a manageable set of dimensions,
    ↳ clustering algorithms like Leiden operate on a graph-based representation of
    ↳ cell-cell relationships.

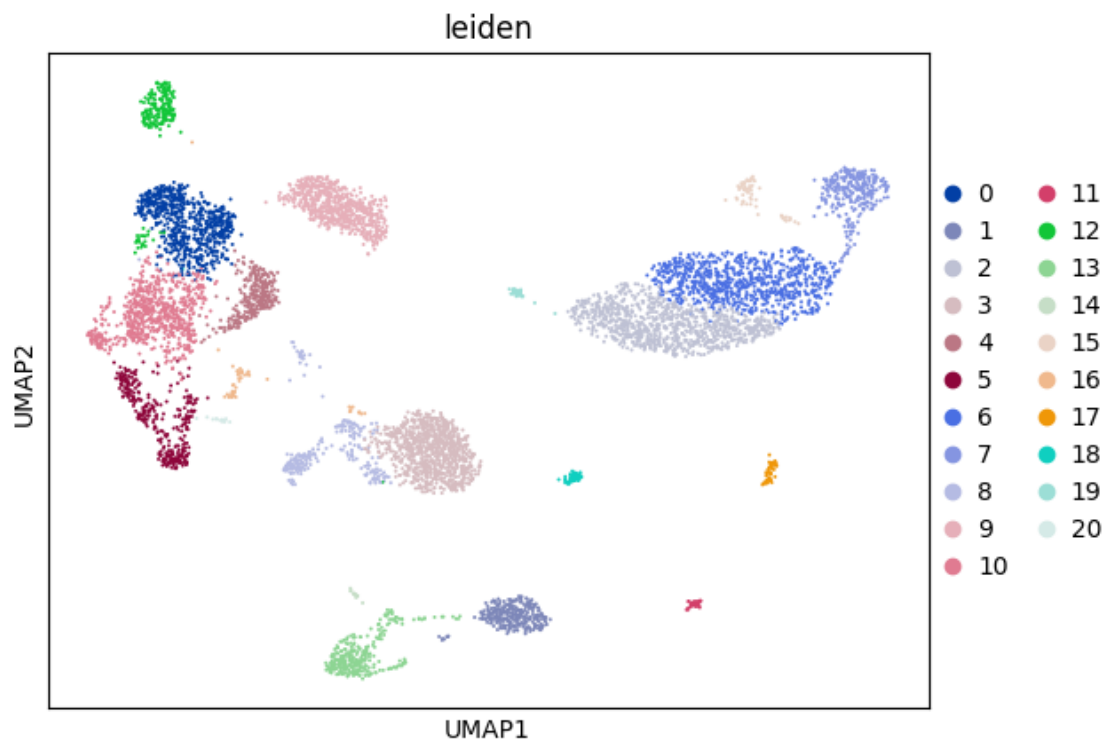
## Usually used for cell type detection

[ ]: # Using the igraph implementation and a fixed number of iterations can be
    ↳ significantly faster, especially for larger datasets
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2)

[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["pct_counts_RIBO"],
    size=8,
)
```

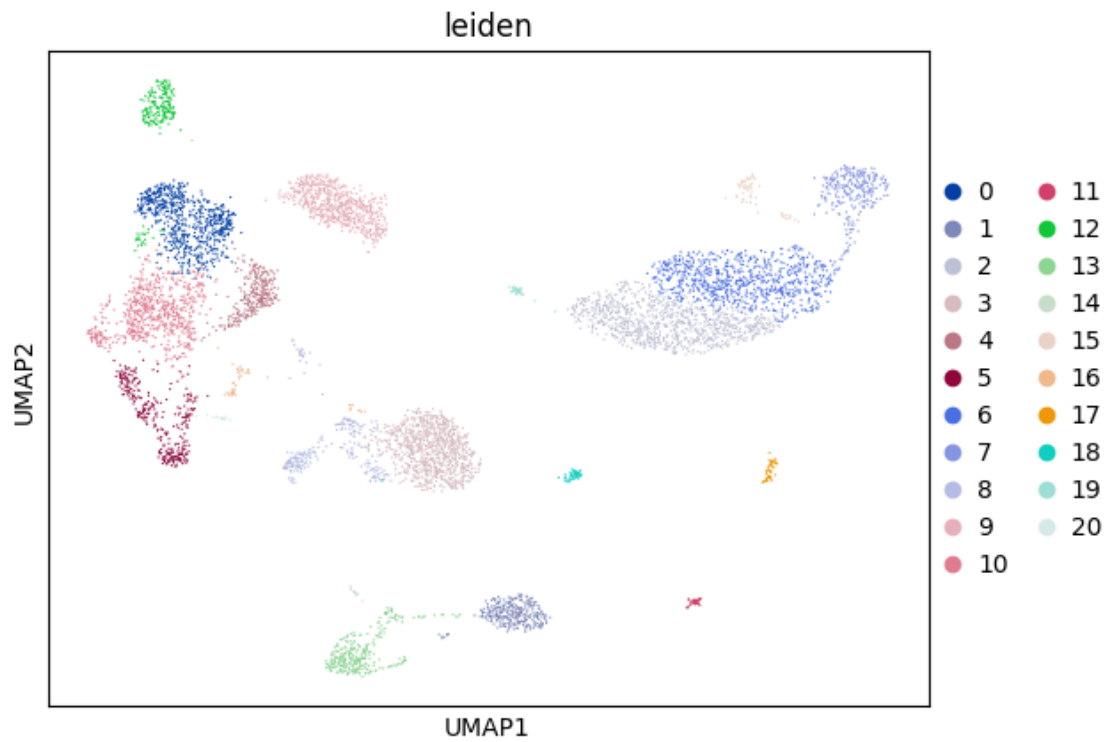


```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=["leiden"],  
    size=8,  
)
```

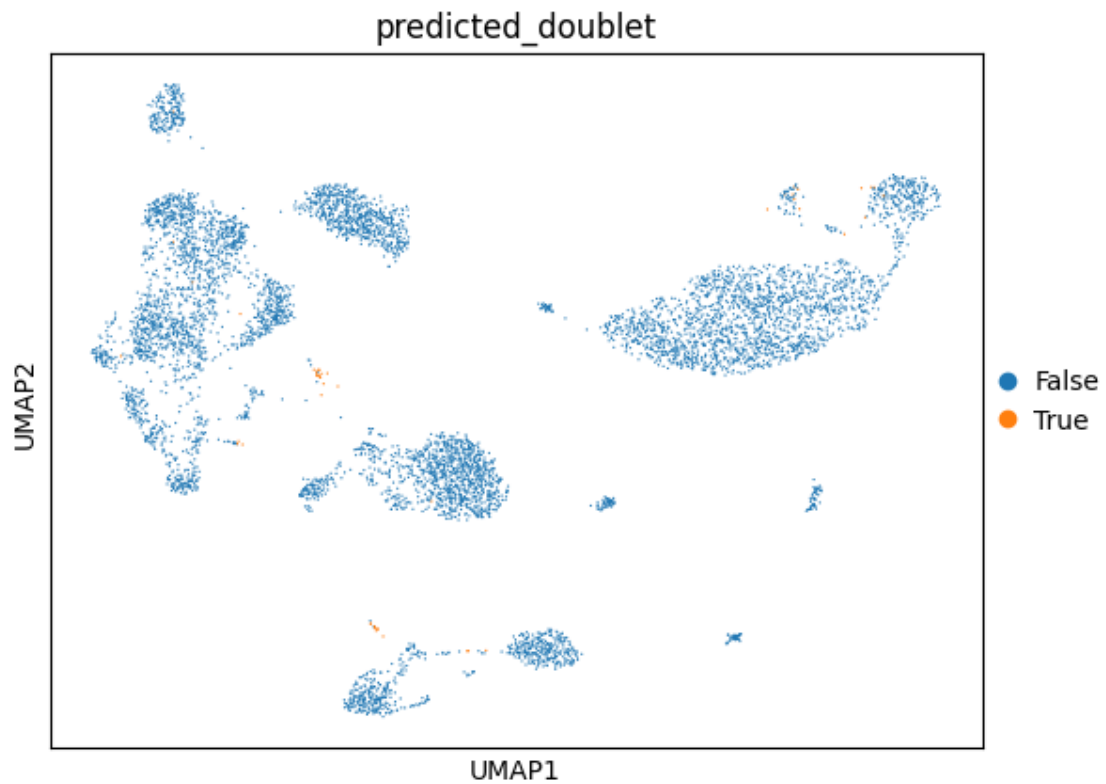


```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["leiden"],
    # increase horizontal space between panels
    wspace=0.5,
    size=3,
    ncols = 1
)
```

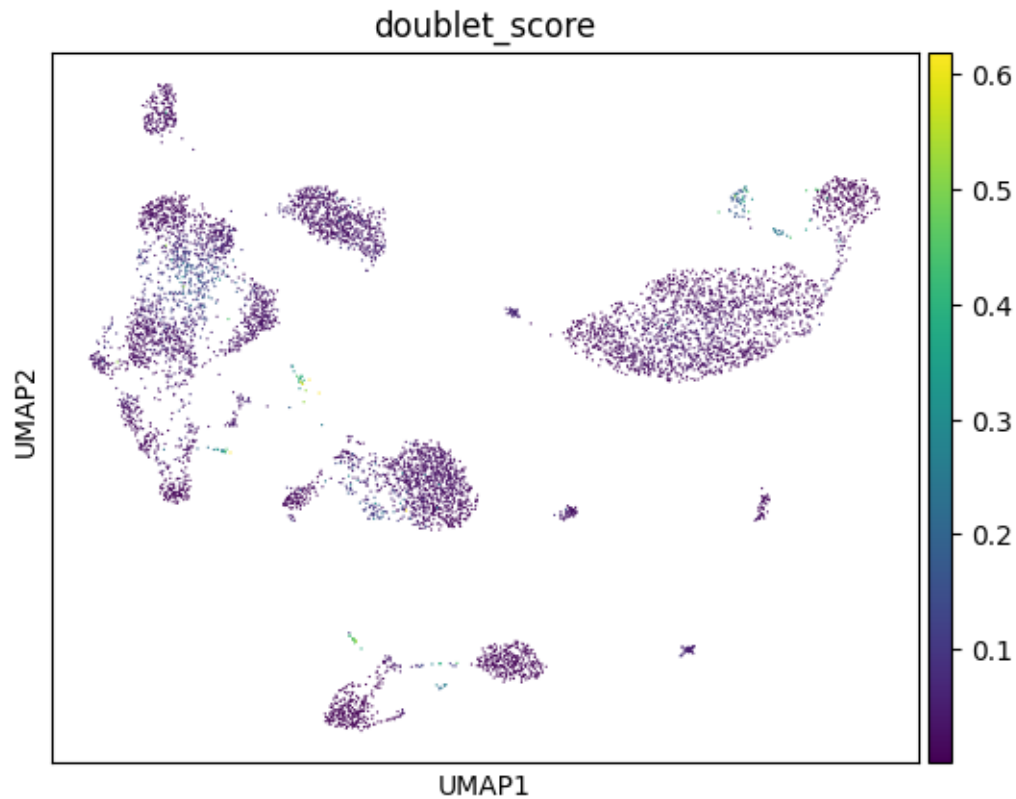




```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=[ "predicted_doublet" ],
    # increase horizontal space between panels
    wspace=0.5,
    size=3,
    ncols = 1
)
```



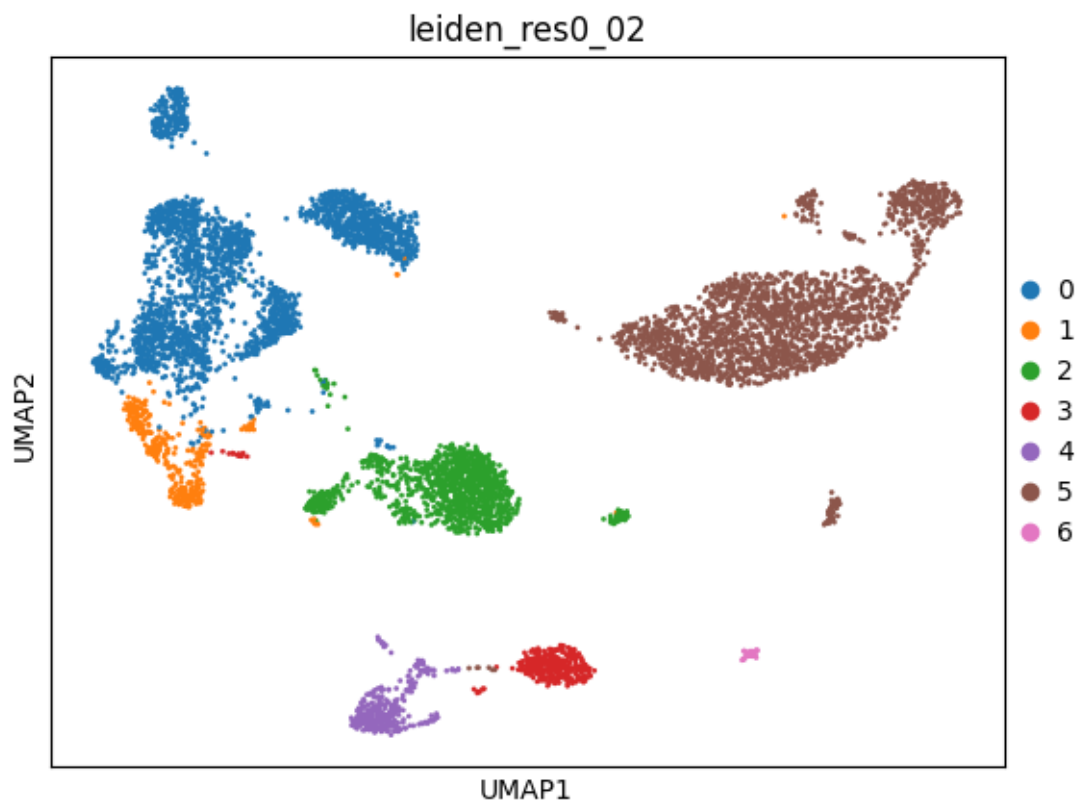
```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=[ "doublet_score"],  
    # increase horizontal space between panels  
    wspace=0.5,  
    size=3,  
    ncols = 1  
)
```



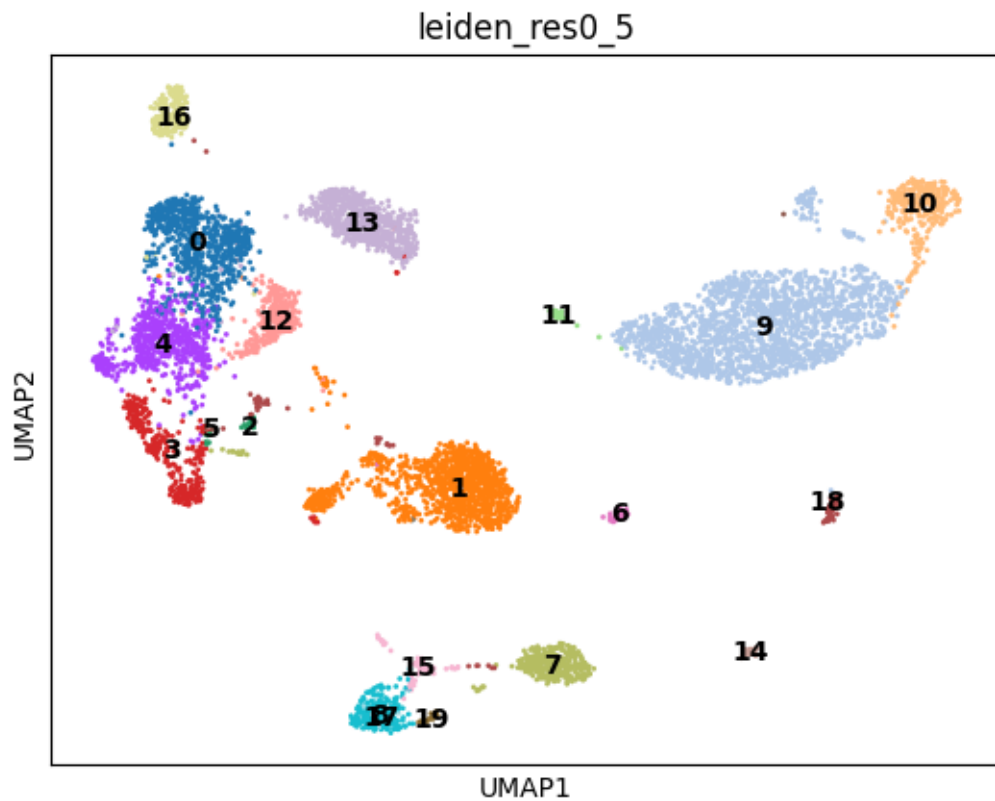
```
[ ]: #Further reclustering
```

```
[ ]: sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2,
    ↪key_added="leiden_res0_02", resolution=0.02)
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2,
    ↪key_added="leiden_res0_5", resolution=0.5)
sc.tl.leiden(fresh_blood_adata, flavor="igraph", n_iterations=2,
    ↪key_added="leiden_res2", resolution=2)
```

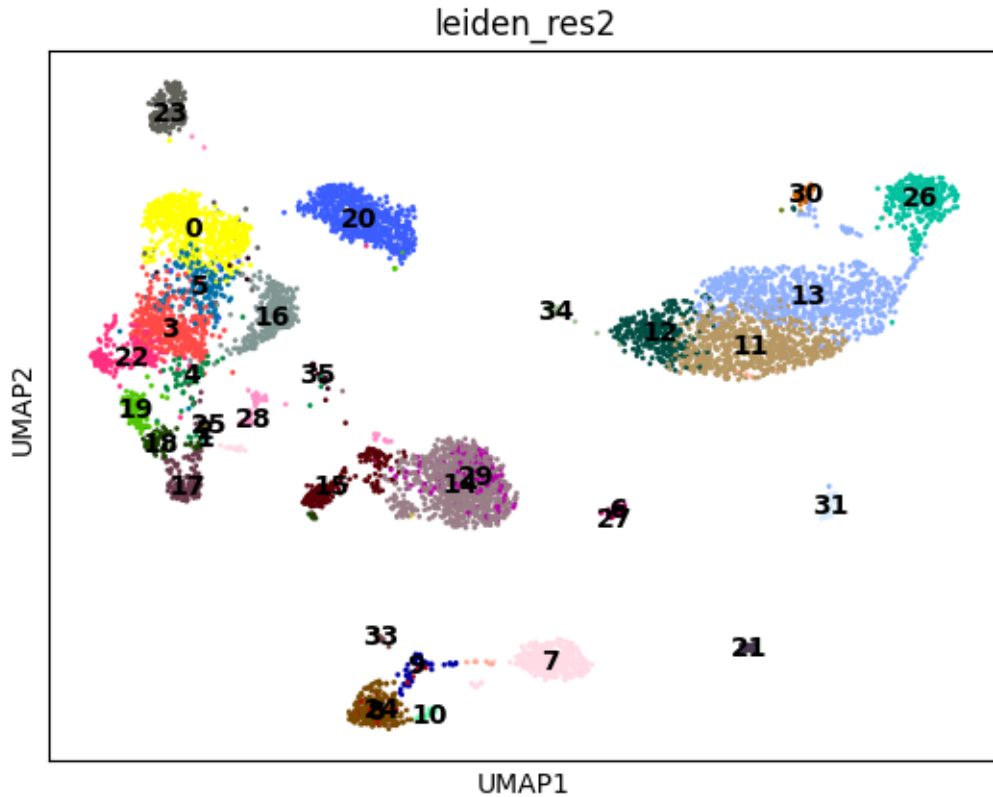
```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["leiden_res0_02"],
    # increase horizontal space between panels
    wspace=0.5,
    size=15,
    ncols = 1
)
```



```
[ ]: sc.pl.umap(  
    fresh_blood_adata,  
    color=["leiden_res0_5"],  
    # increase horizontal space between panels  
    wspace=0.5,  
    size=15,  
    ncol = 1,  
    legend_loc="on data"  
)
```



```
[ ]: sc.pl.umap(
    fresh_blood_adata,
    color=["leiden_res2"],
    # increase horizontal space between panels
    wspace=0.5,
    size=15,
    ncol = 1,
    legend_loc="on data"
)
```



### 1.0.6 Cell Annotation

Cell annotation is the process of assigning biological meaning—like cell type or functional state—to each cluster found after Leiden clustering.

Traditionally, this relies on manual marker gene inspection: you identify top genes per cluster and match them to known markers. But tools like Decoupler enable a more systematic and data-driven approach.

Decoupler is a framework for gene set activity inference. Instead of labeling clusters by single markers, it estimates the activity of predefined pathways, transcription factors, or cell-type signatures from known databases (e.g., MSigDB, PROGENy, DoRothEA).

In practice:

- You provide your normalized expression matrix (adata).
- You load gene sets representing known biological programs or cell-type signatures.
- Decoupler calculates an activity score per cell or cluster using methods like weighted mean, ULM, or AUCell.
- You interpret those activities to annotate clusters automatically or semi-automatically.

```
[ ]: import decoupler as dc
```

```
[ ]: # Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="human")

# Keep canonical cell type markers alone
markers = markers[markers["canonical_marker"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

# Format because dc only accepts cell_type and genesymbol

markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()
```

```
[ ]:           source  target

2      Bergmann glia    A2M
4           Mast cells  A4GALT
5  Embryonic stem cells  A7KBS4
8       Pinealocytes   AANAT
9       Astrocytes     AANAT
```

```
[ ]: #load the gene expression matrix into dc

dc.mt.ulm(data=fresh_blood_adata,
          net=markers,
          tmin = 3)
```

```
[ ]: #retrieve the score for each cell type

score = dc.pp.get_obsm(fresh_blood_adata, key="score_ulm")
score
```

```
[ ]: AnnData object with n_obs × n_vars = 13439 × 97
      obs: 'n_genes_by_counts', 'log1p_n_genes_by_counts', 'total_counts',
          'log1p_total_counts', 'pct_counts_in_top_50_genes',
          'pct_counts_in_top_100_genes', 'pct_counts_in_top_200_genes',
          'pct_counts_in_top_500_genes', 'total_counts_MT', 'log1p_total_counts_MT',
          'pct_counts_MT', 'total_counts_RIBO', 'log1p_total_counts_RIBO',
          'pct_counts_RIBO', 'total_counts_HB', 'log1p_total_counts_HB', 'pct_counts_HB',
          'n_genes', 'doublet_score', 'predicted_doublet', 'leiden', 'leiden_res0_02',
```

```

'leiden_res0_5', 'leiden_res2'
  uns: 'scrublet', 'log1p', 'hvg', 'pca', 'neighbors', 'umap', 'leiden',
'leiden_colors', 'predicted_doublet_colors', 'leiden_res0_02', 'leiden_res0_5',
'leiden_res2', 'leiden_res0_02_colors', 'leiden_res0_5_colors',
'leiden_res2_colors'
  obsm: 'X_pca', 'X_umap', 'score_ulm', 'padj_ulm'

```

```

[ ]: #preview the data
fresh_blood_adata.obsm["score_ulm"].head()

```

```

[ ]:
Acinar cells  Adipocytes  Alpha cells  \
AAACCAAAGAACAGAC-1  -1.024520  -1.157857  -1.122553
AAACCAAAGCAAGATA-1  -0.851239  -1.066545   1.392907
AAACCAAAGCTGGTTA-1  -0.972659  -0.605491   1.733040
AAACCAGCACCTAACG-1  -0.975911  -2.175951   1.169892
AAACCAGCACGCGTTA-1  -0.365460  -1.407562   1.452857

Alveolar macrophages  Astrocytes  B cells  \
AAACCAAAGAACAGAC-1      3.167727   1.788504  0.051880
AAACCAAAGCAAGATA-1     -0.160084   1.674194  1.649956
AAACCAAAGCTGGTTA-1      1.533731   1.245965  1.393795
AAACCAGCACCTAACG-1      1.881583   2.208534 -0.181091
AAACCAGCACGCGTTA-1      1.975010   1.530813  1.212177

B cells memory  B cells naive  Basophils  Bergmann glia  \
AAACCAAAGAACAGAC-1  -1.991232  -1.919460  0.996606   2.561394
AAACCAAAGCAAGATA-1  -0.657855   0.215860  0.280422  -0.392360
AAACCAAAGCTGGTTA-1  -1.627565  -0.710262 -0.071197   0.367371
AAACCAGCACCTAACG-1  -2.086519  -1.618377  1.766395   2.600667
AAACCAGCACGCGTTA-1  -0.524691  -0.725068 -0.173370   1.918253

...  Sebocytes  Sertoli cells  Smooth muscle cells  \
AAACCAAAGAACAGAC-1  ... -0.970895  -0.943990           0.159336
AAACCAAAGCAAGATA-1  ... -1.204066  -1.446825          -1.446825
AAACCAAAGCTGGTTA-1  ... -0.641492   0.093290          -1.218025
AAACCAGCACCTAACG-1  ...  0.863481  -2.325356          -0.313729
AAACCAGCACGCGTTA-1  ...  0.613889  -1.642448          -1.796702

T cells  T follicular helper cells  T helper cells  \
AAACCAAAGAACAGAC-1  11.967801          -0.348311   3.556622
AAACCAAAGCAAGATA-1   6.544548          -0.851239   1.153128
AAACCAAAGCTGGTTA-1  11.277543          -0.972659   1.850920
AAACCAGCACCTAACG-1   9.256290          -0.975911   3.394022
AAACCAGCACGCGTTA-1  11.714664          -0.365460   2.828105

T regulatory cells  Tanycytes  Thymocytes  Tuft cells
AAACCAAAGAACAGAC-1  -0.233372  -1.024520   3.257220   2.539718

```



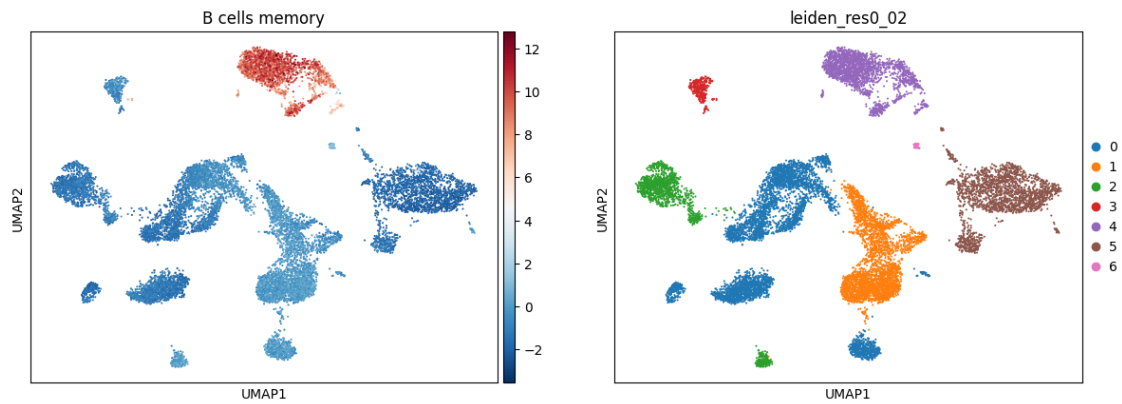
|                    |           |           |          |          |
|--------------------|-----------|-----------|----------|----------|
| AAACCAAAGCAAGATA-1 | -0.715220 | -0.160084 | 1.384593 | 1.146025 |
| AAACCAAAGCTGGTTA-1 | -1.375830 | 0.065565  | 1.963522 | 2.093752 |
| AAACCAGCACCTAACG-1 | -1.380429 | -0.975911 | 2.632058 | 3.212239 |
| AAACCAGCACGCGTTA-1 | -1.366579 | -0.365460 | 3.798206 | 1.923412 |

[5 rows x 97 columns]

```
[ ]: fresh_blood_adata.obsm["score_ulm"].columns
```

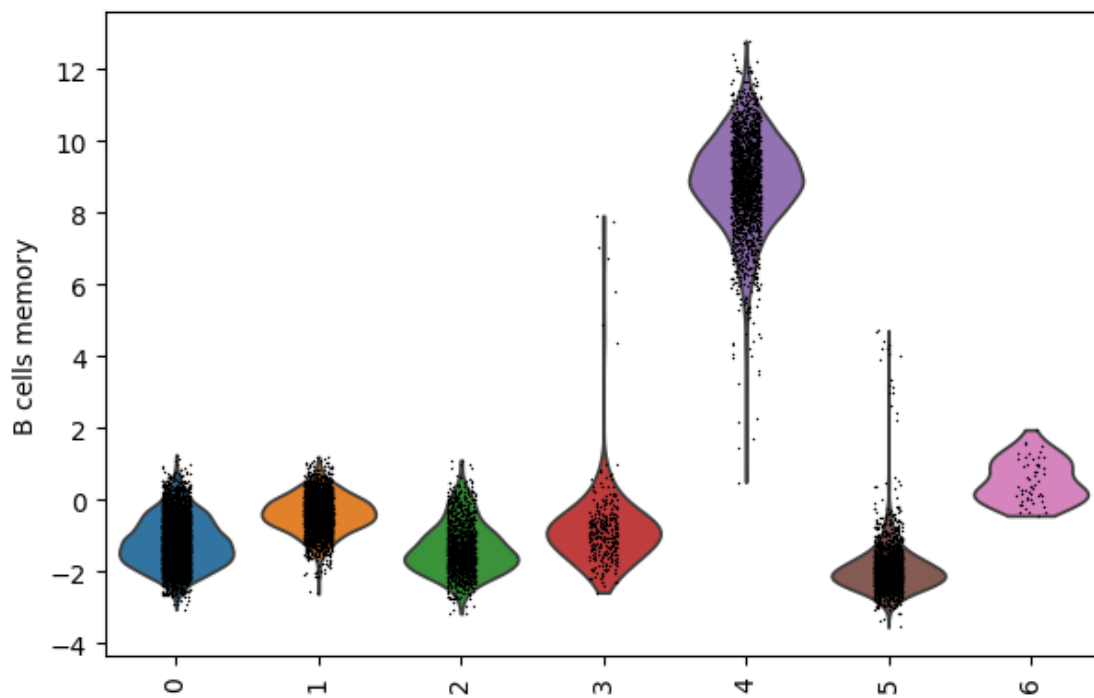
```
[ ]: Index(['Acinar cells', 'Adipocytes', 'Alpha cells', 'Alveolar macrophages',
'Astrocytes', 'B cells', 'B cells memory', 'B cells naive', 'Basophils',
'Bergmann glia', 'Beta cells', 'Cardiomyocytes', 'Cholangiocytes',
'Chondrocytes', 'Chromaffin cells', 'Clara cells', 'Crypt cells',
'Delta cells', 'Dendritic cells', 'Ductal cells',
'Embryonic stem cells', 'Endothelial cells', 'Enterocytes',
'Enteroendocrine cells', 'Eosinophils', 'Ependymal cells',
'Epsilon cells', 'Erythroblasts',
'Erythroid-like and erythroid precursor cells', 'Fibroblasts',
'Gamma (PP) cells', 'Gamma delta T cells', 'Gastric chief cells',
'Germ cells', 'Glomus cells', 'Hematopoietic stem cells', 'Hepatocytes',
'Immature neurons', 'Intercalated cells', 'Keratinocytes',
'Kupffer cells', 'Langerhans cells', 'Leydig cells',
'Loop of Henle cells', 'Luminal epithelial cells', 'Macrophages',
'Mammary epithelial cells', 'Mast cells', 'Megakaryocytes',
'Melanocytes', 'Merkel cells', 'Mesangial cells', 'Microfold cells',
'Microglia', 'Monocytes', 'Myoblasts', 'Müller cells', 'NK cells',
'Natural killer T cells', 'Neural stem/precursor cells', 'Neuroblasts',
'Neurons', 'Neutrophils', 'Nuocytes',
'Oligodendrocyte progenitor cells', 'Oligodendrocytes', 'Osteoblasts',
'Osteoclasts', 'Pancreatic progenitor cells', 'Paneth cells',
'Pericytes', 'Pinealocytes', 'Plasma cells',
'Plasmacytoid dendritic cells', 'Platelets', 'Pluripotent stem cells',
'Podocytes', 'Principal cells', 'Proximal tubule cells',
'Pulmonary alveolar type I cells', 'Pulmonary alveolar type II cells',
'Pyramidal cells', 'Red pulp macrophages', 'Reticulocytes',
'Satellite cells', 'Satellite glial cells', 'Schwann cells',
'Sebocytes', 'Sertoli cells', 'Smooth muscle cells', 'T cells',
'T follicular helper cells', 'T helper cells', 'T regulatory cells',
'Tanocytes', 'Thymocytes', 'Tuft cells'],
dtype='object')
```

```
[ ]: sc.pl.umap(score, color=["B cells memory", "leiden_res0_02"], cmap="RdBu_r")
```

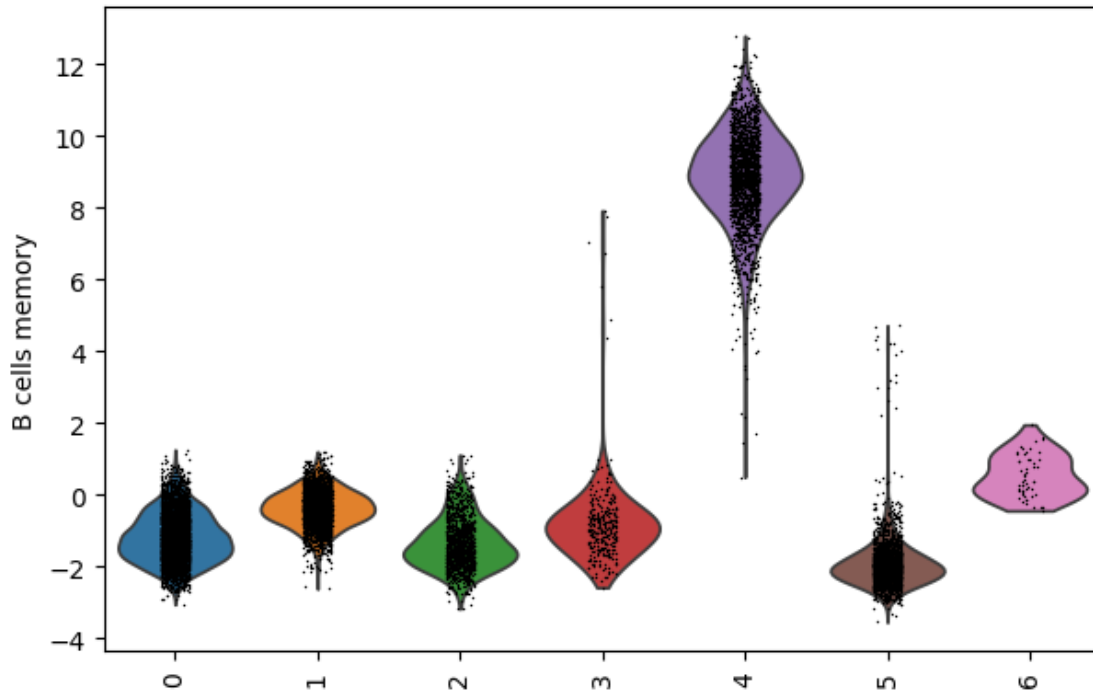


```
[ ]: import seaborn as sns
```

```
[ ]: sc.pl.violin(score, keys=["B cells memory"], groupby="leiden_res0_02",
    ↪rotation=90)
```



```
[ ]: sc.pl.violin(score, keys=["B cells memory"], groupby="leiden_res0_02",
    ↪rotation=90)
```



```
[ ]: #. Now let's know what each of the 7 clusters mean
```

```
[ ]: #rank genes
fresh_blood_adata_rank = dc.tl.rankby_group(score, groupby="leiden_res0_02",
↪reference="rest", method="t-test_overestim_var")
fresh_blood_adata_rank = fresh_blood_adata_rank[fresh_blood_adata_rank["stat"]
↪> 0]
fresh_blood_adata_rank.head()
```

```
[ ]:  group reference      name      stat  meanchange  pval  padj
0      0      rest      T cells  87.319301    6.030998  0.0   0.0
1      0      rest      NK cells  76.073701    6.432556  0.0   0.0
2      0      rest  Gamma delta T cells  62.249197    5.416916  0.0   0.0
3      0      rest      Thymocytes  60.394609    1.450974  0.0   0.0
4      0      rest  Natural killer T cells  39.891940    1.380208  0.0   0.0
```

```
[ ]: cluster_annotations = fresh_blood_adata_rank[fresh_blood_adata_rank["stat"] >
↪0].groupby("group").head(1).set_index("group")["name"].to_dict()
```

```
[ ]: cluster_annotations
```

```
[ ]: {'0': 'T cells',
      '1': 'Nuocytes',
      '2': 'Natural killer T cells',
```

```
'3': 'Mast cells',
'4': 'B cells memory',
'5': 'Macrophages',
'6': 'Plasmacytoid dendritic cells'}
```

```
[ ]: fresh_blood_adata.obs['cell_type'] = fresh_blood_adata.obs['leiden_res0_02'].
      ↪map(cluster_annotations)
```

```
[ ]: # Example of how to subset for multiple genes in the 'source' column
available_genes = set(fresh_blood_adata.var_names)

b_cell_markers = markers[markers['source'].isin(['B cells memory'])]['target']
b_cell_markers = b_cell_markers[b_cell_markers.isin(available_genes)]

nk_cell_markers = markers[markers['source'].isin(['Natural killer T_
      ↪cells'])]['target']
nk_cell_markers = nk_cell_markers[nk_cell_markers.isin(available_genes)]

t_cells_markers = markers[markers['source'].isin(['T cells'])]['target']
t_cells_markers = t_cells_markers[t_cells_markers.isin(available_genes)]

#display(b_cell_markers)
```

### 1.0.7 Other ways to visualize the cell types

```
[ ]: marker_genes_dict = {
      "B cells": b_cell_markers.head().tolist(),
      "NK cells": nk_cell_markers.head().tolist(),
      "T cells": t_cells_markers.head().tolist()
    }
```

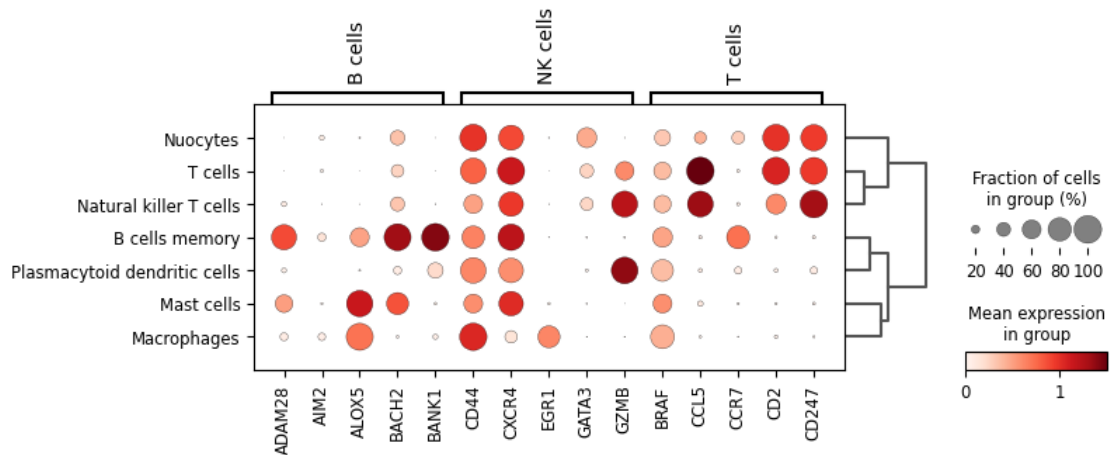
```
[ ]: sc.pl.dotplot(fresh_blood_adata, marker_genes_dict, "cell_type",
      ↪dendrogram=True)
```

WARNING: dendrogram data not found (using key=dendrogram\_cell\_type). Running `sc.tl.dendrogram` with default parameters. For fine tuning it is recommended to run `sc.tl.dendrogram` independently.

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.

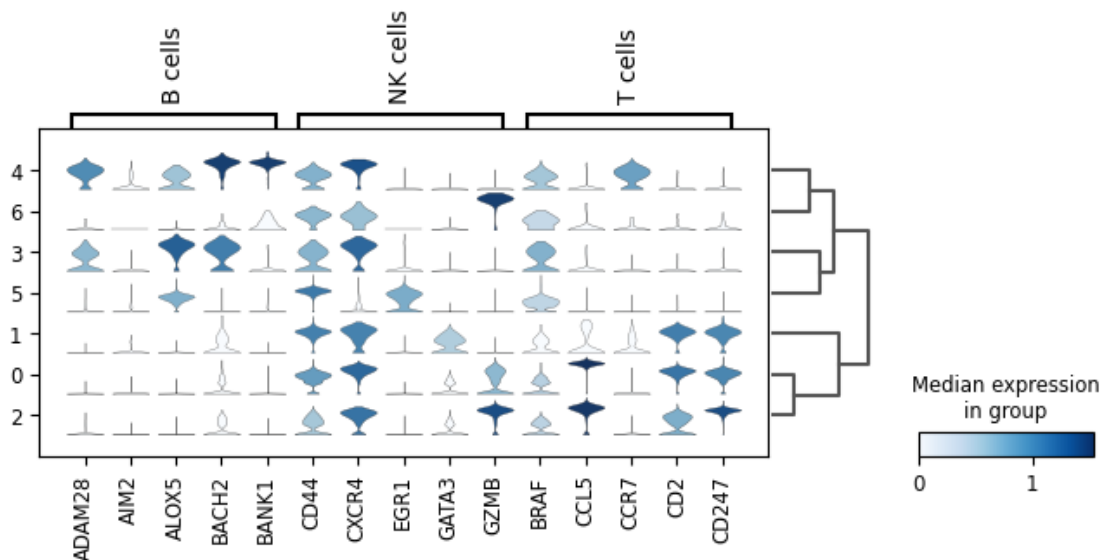
categories: T cells, Nuocytes, Natural killer T cells, etc.

var\_group\_labels: B cells, NK cells, T cells



```
[ ]: sc.pl.stacked_violin(
    fresh_blood_adata, marker_genes_dict, groupby="leiden_res0_02",
    dendrogram=True
)
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.  
categories: 0, 1, 2, etc.  
var\_group\_labels: B cells, NK cells, T cells

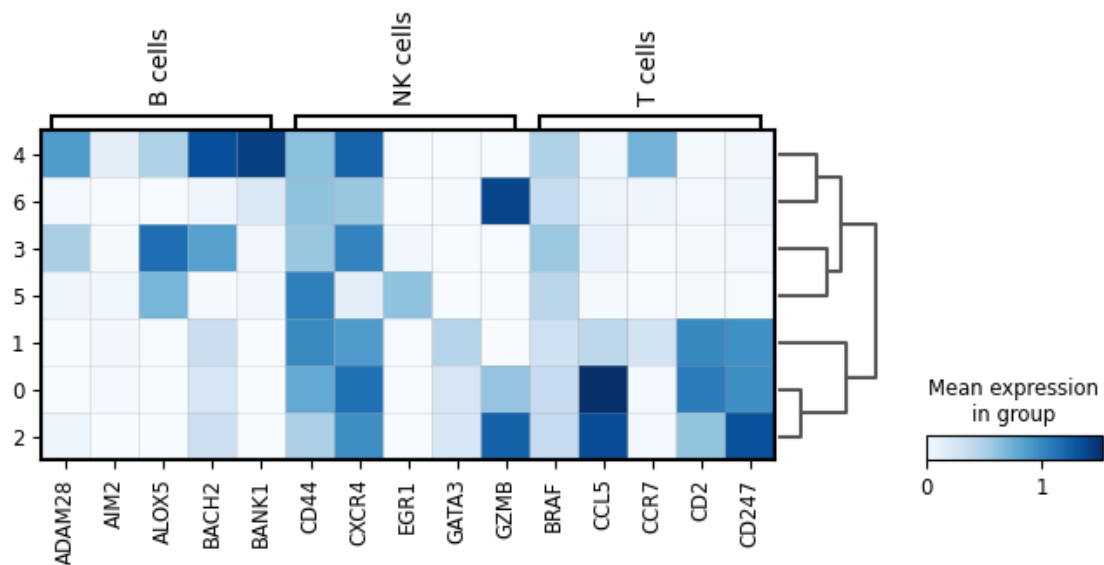


```
[ ]: sc.pl.matrixplot(
    fresh_blood_adata,
    marker_genes_dict,
    "leiden_res0_02",
    dendrogram=True,
    cmap="Blues",
)
```

WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.

categories: 0, 1, 2, etc.

var\_group\_labels: B cells, NK cells, T cells

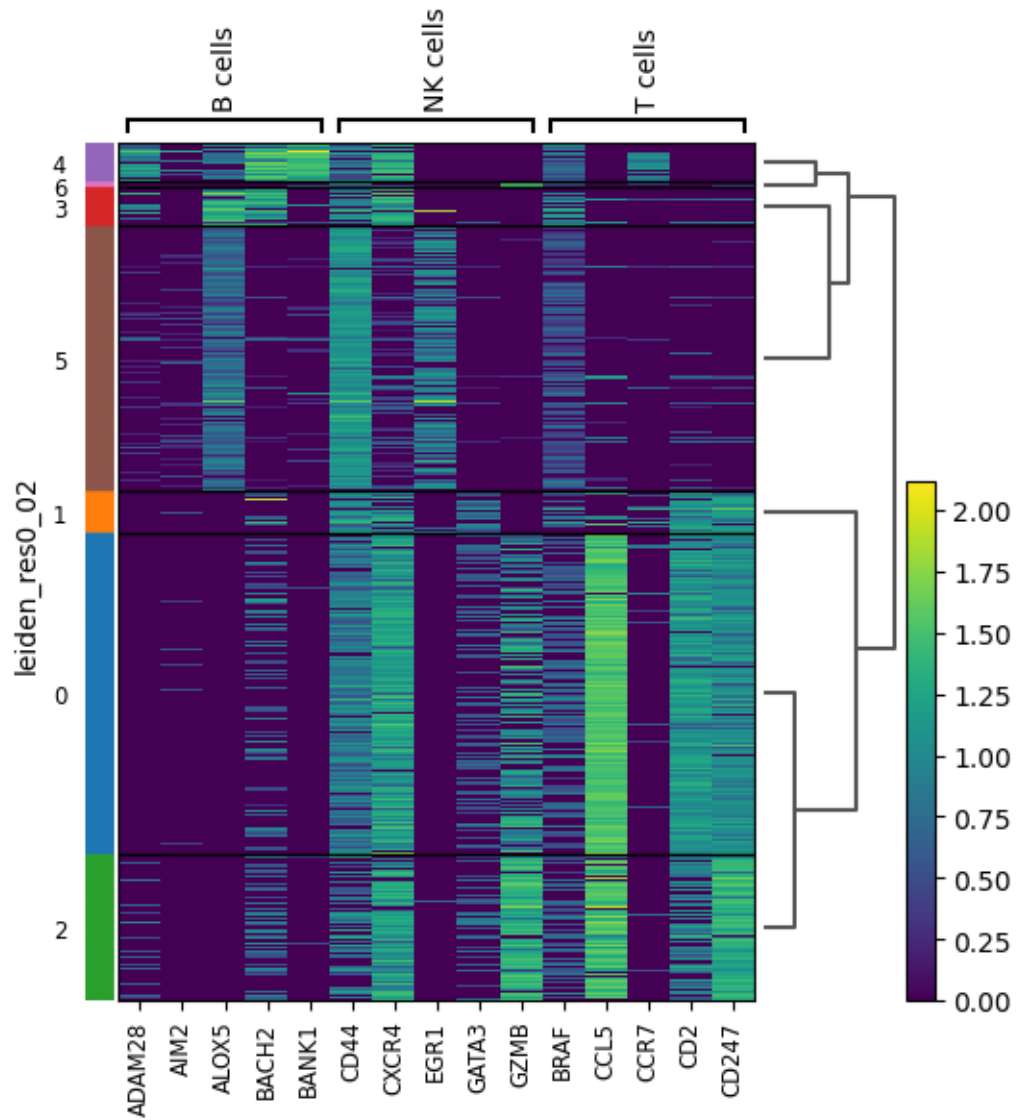


```
[ ]: sc.pl.heatmap(
    fresh_blood_adata, marker_genes_dict, groupby="leiden_res0_02",
    cmap="viridis", dendrogram=True
)
```

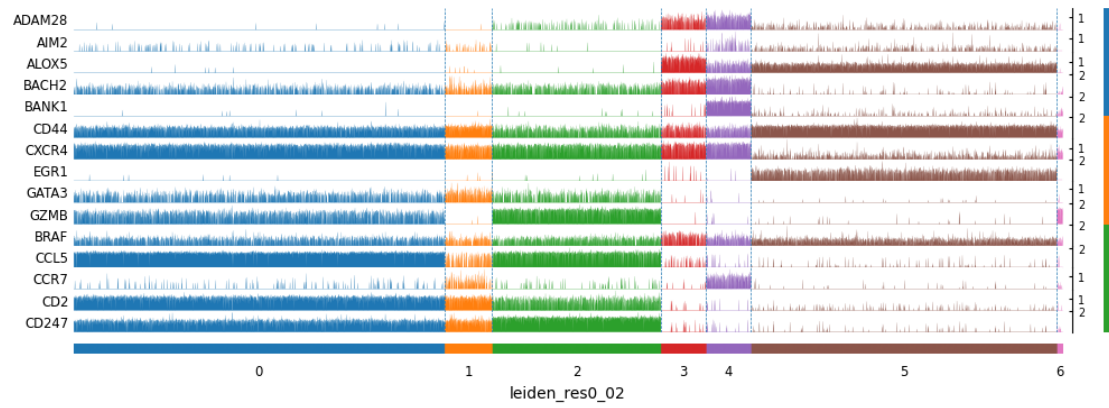
WARNING: Groups are not reordered because the `groupby` categories and the `var\_group\_labels` are different.

categories: 0, 1, 2, etc.

var\_group\_labels: B cells, NK cells, T cells



```
[ ]: # @title Using genome tracks
sc.pl.tracksplot(fresh_blood_adata, marker_genes_dict,
↳groupby="leiden_res0_02", dendrogram=False)
```



[ ]: